

# Score Forecasts

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## Load in truth data and forecasts

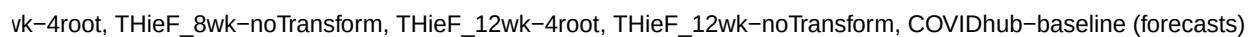
We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

## Plot Forecasts

We plot forecasts for three locations (US National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 10/31/21.

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## [1] 18603
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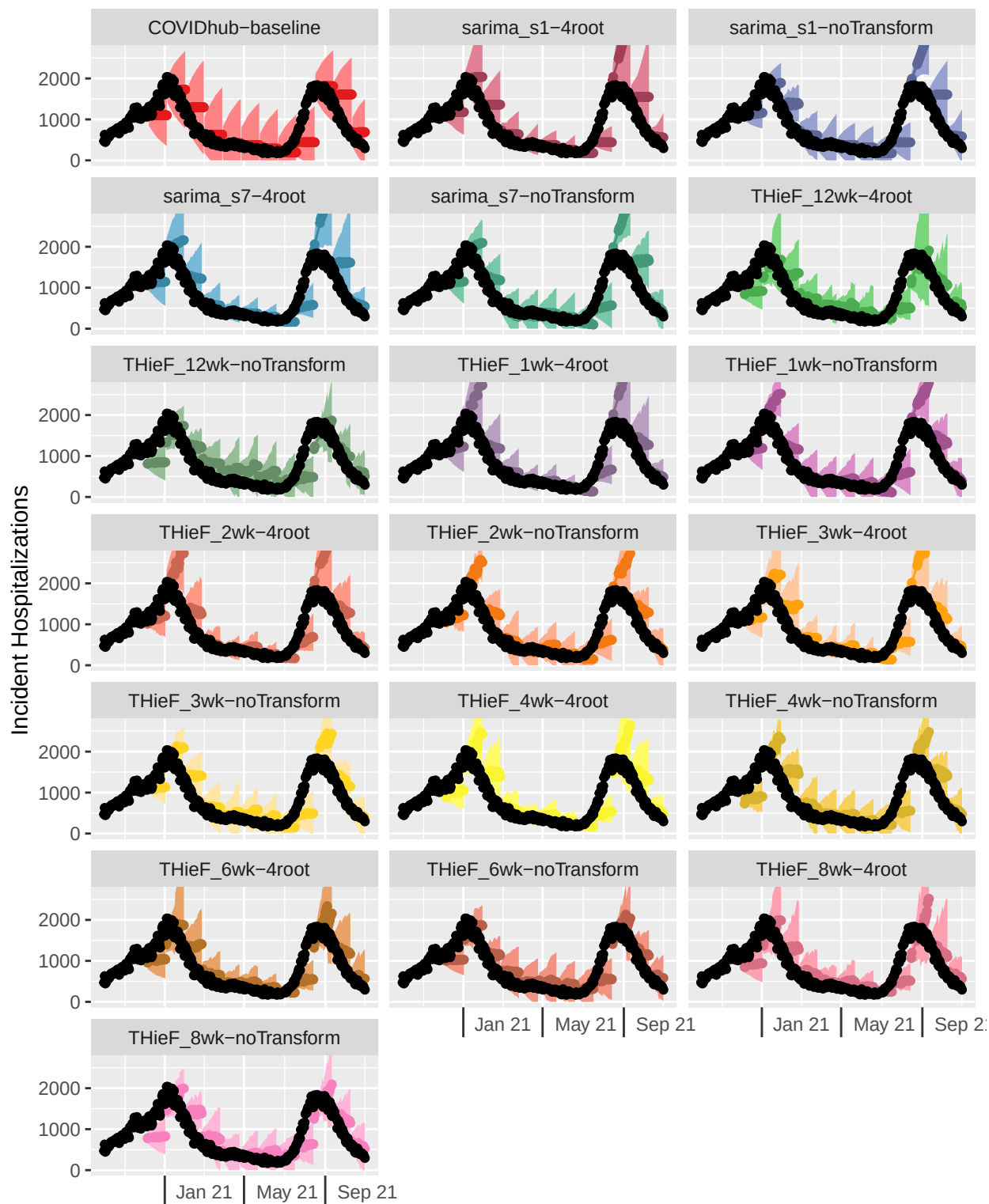
Selected forecast date(s): 2021-01-29, 2021-02-26, 2021-03-27, 2021-04-24, 2021-05-23, 2021-06-27, 2021-07-27, 2021-08-31, 2021-09-28, 2021-10-29, 2021-11-26, 2021-12-31



## Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Texas

Selected forecast date(s): 2020-12-05, 2021-01-02, 2021-01-29, 2021-02-26, 2021

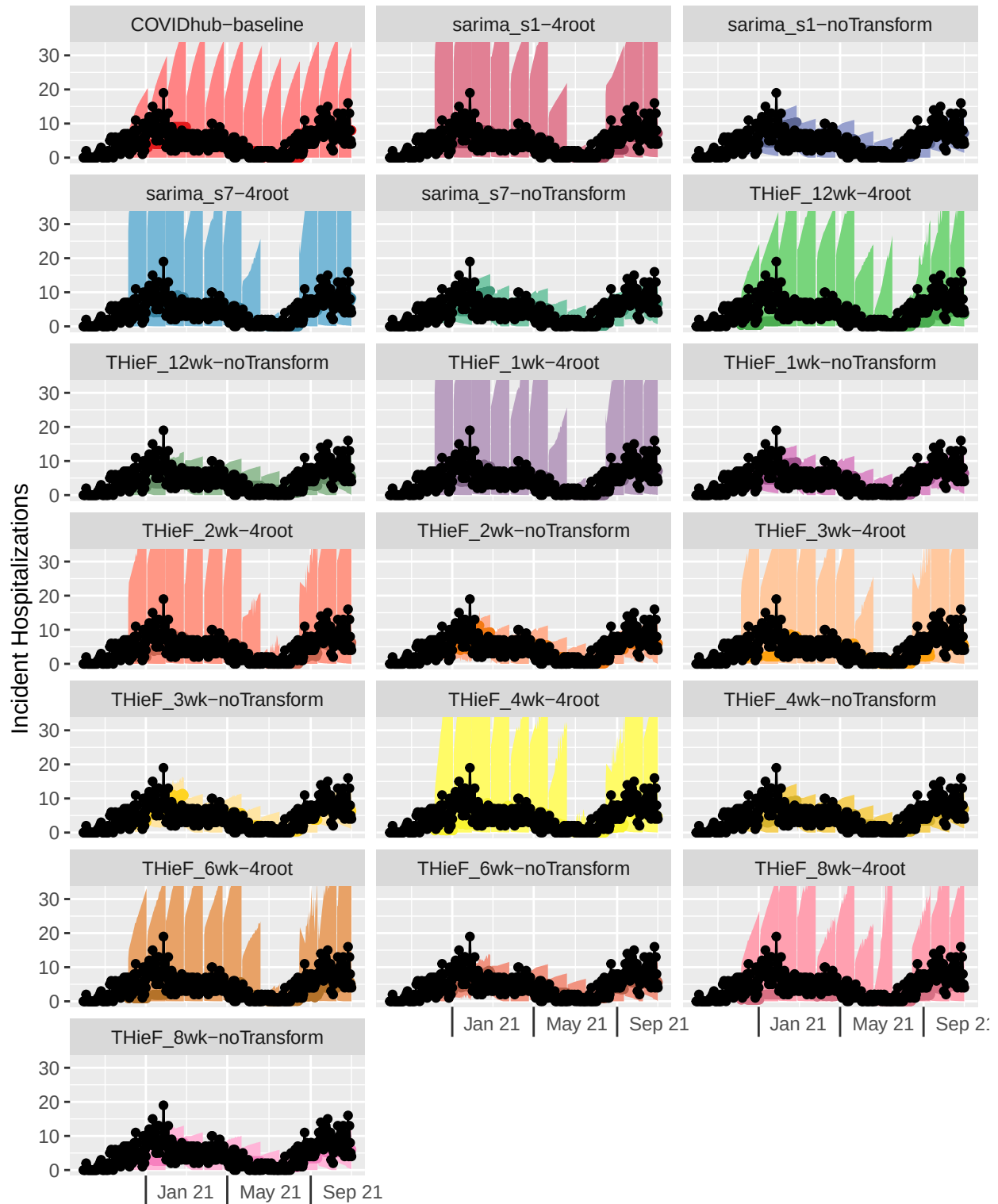


vk-4root, THieF\_8wk-noTransform, THieF\_12wk-4root, THieF\_12wk-noTransform, COVIDhub-baseline (forecasts)

## Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Vermont

Selected forecast date(s): 2020-12-05, 2021-01-02, 2021-01-29, 2021-02-26, 2021-03-26



vk-4root, THieF\_8wk-noTransform, THieF\_12wk-4root, THieF\_12wk-noTransform, COVIDhub-baseline (forecasts)

The plots don't show much difference between the 6 models (although this is partially due to the fact that the y-scales are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

## Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

## Overall metrics

| model                  | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|--------|--------|--------|--------|-------|-------|
| THieF_6wk-noTransform  | 1071.3 | 1723.5 | 0.344  | 0.924  | 0.828 | 0.846 |
| THieF_1wk-noTransform  | 1123.0 | 1766.7 | 0.450  | 0.886  | 0.868 | 0.868 |
| THieF_6wk-4root        | 1133.3 | 1744.0 | 0.311  | 0.755  | 0.876 | 0.856 |
| THieF_2wk-4root        | 1164.9 | 1867.0 | 0.228  | 0.860  | 0.900 | 0.917 |
| THieF_1wk-4root        | 1175.4 | 1908.8 | 0.280  | 0.866  | 0.908 | 0.937 |
| THieF_8wk-noTransform  | 1246.3 | 1948.7 | 0.371  | 0.856  | 0.963 | 0.957 |
| THieF_2wk-noTransform  | 1251.3 | 1909.6 | 0.387  | 0.836  | 0.967 | 0.938 |
| COVIDhub-baseline      | 1294.1 | 2036.2 | 0.622  | 0.986  | 1.000 | 1.000 |
| THieF_8wk-4root        | 1297.7 | 1959.8 | 0.233  | 0.689  | 1.003 | 0.962 |
| THieF_3wk-noTransform  | 1316.6 | 1953.5 | 0.368  | 0.788  | 1.017 | 0.959 |
| THieF_3wk-4root        | 1329.9 | 2096.3 | 0.242  | 0.805  | 1.028 | 1.030 |
| THieF_4wk-noTransform  | 1342.8 | 2004.8 | 0.331  | 0.776  | 1.038 | 0.985 |
| THieF_4wk-4root        | 1348.7 | 2028.1 | 0.259  | 0.695  | 1.042 | 0.996 |
| sarima_s7-noTransform  | 1401.9 | 2124.6 | 0.382  | 0.819  | 1.083 | 1.043 |
| sarima_s1-noTransform  | 1419.3 | 2110.1 | 0.365  | 0.800  | 1.097 | 1.036 |
| sarima_s1-4root        | 1482.8 | 2169.6 | 0.253  | 0.750  | 1.146 | 1.066 |
| sarima_s7-4root        | 1537.6 | 2398.7 | 0.208  | 0.770  | 1.188 | 1.178 |
| THieF_12wk-noTransform | 1548.1 | 2378.9 | 0.301  | 0.764  | 1.196 | 1.168 |
| THieF_12wk-4root       | 2334.4 | 2060.9 | 0.320  | 0.650  | 1.804 | 1.012 |

| model                 | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-----------------------|--------|--------|--------|--------|-------|-------|
| THieF_6wk-4root       | 31.653 | 48.108 | 0.456  | 0.908  | 0.882 | 0.963 |
| THieF_6wk-noTransform | 32.256 | 48.674 | 0.540  | 0.917  | 0.899 | 0.974 |
| THieF_1wk-4root       | 32.678 | 50.063 | 0.480  | 0.911  | 0.911 | 1.002 |
| THieF_2wk-4root       | 32.936 | 50.259 | 0.474  | 0.909  | 0.918 | 1.006 |
| THieF_2wk-noTransform | 33.258 | 48.027 | 0.574  | 0.900  | 0.927 | 0.961 |
| THieF_1wk-noTransform | 33.736 | 49.209 | 0.576  | 0.898  | 0.940 | 0.985 |
| THieF_8wk-4root       | 33.844 | 51.142 | 0.428  | 0.887  | 0.943 | 1.023 |
| THieF_4wk-4root       | 34.096 | 51.501 | 0.456  | 0.896  | 0.950 | 1.031 |

| model                  | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|--------|--------|--------|--------|-------|-------|
| THieF_3wk-4root        | 34.622 | 52.708 | 0.460  | 0.899  | 0.965 | 1.055 |
| THieF_8wk-noTransform  | 34.761 | 52.057 | 0.513  | 0.900  | 0.969 | 1.042 |
| THieF_3wk-noTransform  | 34.960 | 52.077 | 0.535  | 0.896  | 0.974 | 1.042 |
| THieF_4wk-noTransform  | 35.633 | 52.819 | 0.529  | 0.891  | 0.993 | 1.057 |
| COVIDhub-baseline      | 35.877 | 49.975 | 0.826  | 0.983  | 1.000 | 1.000 |
| sarima_s7-noTransform  | 36.292 | 51.906 | 0.567  | 0.886  | 1.012 | 1.039 |
| sarima_s7-4root        | 36.346 | 54.549 | 0.465  | 0.898  | 1.013 | 1.092 |
| sarima_s1-4root        | 36.913 | 55.430 | 0.455  | 0.887  | 1.029 | 1.109 |
| THieF_12wk-noTransform | 39.140 | 59.057 | 0.421  | 0.891  | 1.091 | 1.182 |
| sarima_s1-noTransform  | 39.412 | 56.378 | 0.543  | 0.877  | 1.099 | 1.128 |
| THieF_12wk-4root       | 84.321 | 54.493 | 0.408  | 0.864  | 2.350 | 1.090 |

### Horizon week metrics

| model                  | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|------------|--------|--------|--------|--------|-------|-------|
| THieF_1wk-4root        | 1          | 456.3  | 743.1  | 0.332  | 0.906  | 0.675 | 0.740 |
| THieF_1wk-noTransform  | 1          | 468.5  | 713.5  | 0.465  | 0.892  | 0.694 | 0.710 |
| THieF_2wk-4root        | 1          | 481.3  | 799.5  | 0.315  | 0.941  | 0.712 | 0.796 |
| THieF_2wk-noTransform  | 1          | 567.5  | 877.1  | 0.455  | 0.902  | 0.840 | 0.873 |
| sarima_s7-noTransform  | 1          | 583.2  | 872.9  | 0.395  | 0.818  | 0.863 | 0.869 |
| sarima_s7-4root        | 1          | 598.7  | 942.2  | 0.238  | 0.780  | 0.886 | 0.938 |
| THieF_6wk-noTransform  | 1          | 640.2  | 1047.2 | 0.318  | 0.937  | 0.948 | 1.042 |
| THieF_3wk-4root        | 1          | 664.7  | 1057.3 | 0.210  | 0.794  | 0.984 | 1.052 |
| THieF_6wk-4root        | 1          | 672.1  | 1065.7 | 0.308  | 0.766  | 0.995 | 1.061 |
| COVIDhub-baseline      | 1          | 675.5  | 1004.6 | 0.706  | 1.000  | 1.000 | 1.000 |
| sarima_s1-noTransform  | 1          | 680.8  | 1062.5 | 0.500  | 0.909  | 1.008 | 1.058 |
| sarima_s1-4root        | 1          | 698.3  | 1104.2 | 0.360  | 0.885  | 1.034 | 1.099 |
| THieF_3wk-noTransform  | 1          | 716.4  | 1100.0 | 0.346  | 0.846  | 1.060 | 1.095 |
| THieF_4wk-4root        | 1          | 760.9  | 1183.9 | 0.259  | 0.755  | 1.126 | 1.179 |
| THieF_4wk-noTransform  | 1          | 779.3  | 1201.5 | 0.322  | 0.822  | 1.154 | 1.196 |
| THieF_8wk-noTransform  | 1          | 852.5  | 1346.1 | 0.304  | 0.825  | 1.262 | 1.340 |
| THieF_8wk-4root        | 1          | 953.3  | 1454.7 | 0.157  | 0.622  | 1.411 | 1.448 |
| THieF_12wk-4root       | 1          | 1093.1 | 1554.4 | 0.283  | 0.640  | 1.618 | 1.547 |
| THieF_12wk-noTransform | 1          | 1236.0 | 1854.2 | 0.241  | 0.717  | 1.830 | 1.846 |
| THieF_6wk-noTransform  | 2          | 859.8  | 1401.0 | 0.373  | 0.953  | 0.779 | 0.816 |
| THieF_1wk-4root        | 2          | 865.9  | 1406.0 | 0.290  | 0.871  | 0.785 | 0.819 |
| THieF_1wk-noTransform  | 2          | 868.5  | 1339.1 | 0.419  | 0.885  | 0.787 | 0.780 |
| THieF_2wk-4root        | 2          | 885.0  | 1444.3 | 0.247  | 0.882  | 0.802 | 0.841 |
| THieF_6wk-4root        | 2          | 900.3  | 1412.5 | 0.344  | 0.803  | 0.816 | 0.823 |
| THieF_2wk-noTransform  | 2          | 1006.0 | 1535.6 | 0.380  | 0.835  | 0.912 | 0.895 |
| THieF_3wk-4root        | 2          | 1049.5 | 1660.2 | 0.251  | 0.810  | 0.951 | 0.967 |
| THieF_3wk-noTransform  | 2          | 1056.7 | 1575.8 | 0.412  | 0.792  | 0.958 | 0.918 |
| THieF_8wk-noTransform  | 2          | 1075.3 | 1729.9 | 0.391  | 0.885  | 0.975 | 1.008 |
| sarima_s7-noTransform  | 2          | 1079.1 | 1628.4 | 0.398  | 0.828  | 0.978 | 0.949 |
| COVIDhub-baseline      | 2          | 1103.4 | 1716.6 | 0.664  | 0.997  | 1.000 | 1.000 |
| sarima_s7-4root        | 2          | 1125.0 | 1762.7 | 0.215  | 0.789  | 1.020 | 1.027 |
| THieF_8wk-4root        | 2          | 1128.2 | 1741.6 | 0.247  | 0.717  | 1.022 | 1.015 |
| THieF_4wk-4root        | 2          | 1133.2 | 1707.6 | 0.323  | 0.724  | 1.027 | 0.995 |
| THieF_4wk-noTransform  | 2          | 1149.8 | 1701.7 | 0.373  | 0.796  | 1.042 | 0.991 |
| sarima_s1-noTransform  | 2          | 1192.8 | 1816.5 | 0.358  | 0.835  | 1.081 | 1.058 |

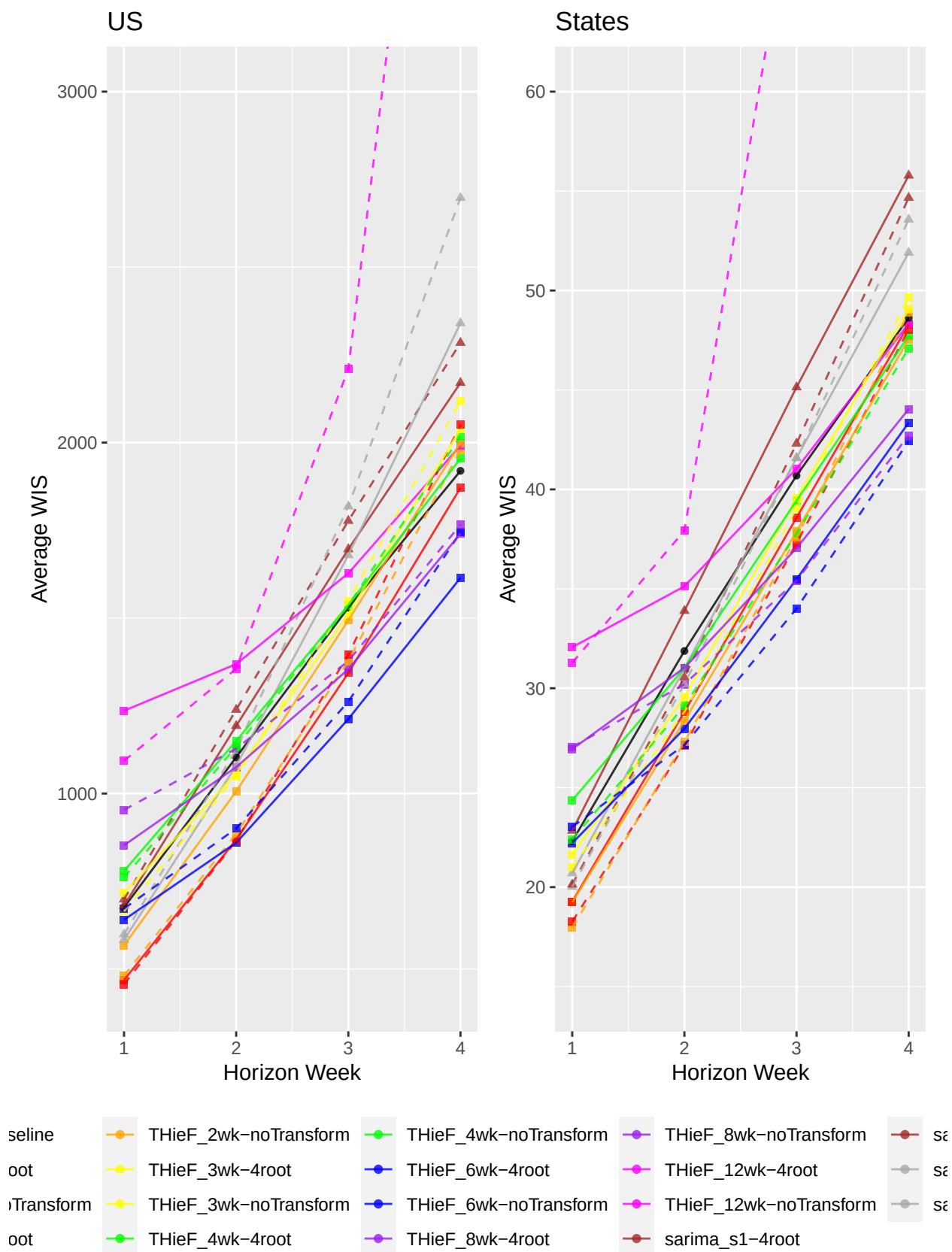
| model                  | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|------------|--------|--------|--------|--------|-------|-------|
| sarima_s1-4root        | 2          | 1238.8 | 1871.1 | 0.229  | 0.756  | 1.123 | 1.090 |
| THieF_12wk-4root       | 2          | 1355.8 | 1790.0 | 0.344  | 0.713  | 1.229 | 1.043 |
| THieF_12wk-noTransform | 2          | 1368.9 | 2149.0 | 0.262  | 0.792  | 1.241 | 1.252 |
| THieF_6wk-noTransform  | 3          | 1212.2 | 1959.4 | 0.368  | 0.919  | 0.792 | 0.805 |
| THieF_6wk-4root        | 3          | 1262.0 | 1942.3 | 0.342  | 0.746  | 0.825 | 0.798 |
| THieF_1wk-noTransform  | 3          | 1343.5 | 2110.2 | 0.449  | 0.875  | 0.878 | 0.867 |
| THieF_8wk-noTransform  | 3          | 1354.1 | 2109.5 | 0.423  | 0.875  | 0.885 | 0.867 |
| THieF_8wk-4root        | 3          | 1377.6 | 2074.5 | 0.276  | 0.710  | 0.901 | 0.853 |
| THieF_2wk-4root        | 3          | 1377.7 | 2207.4 | 0.184  | 0.835  | 0.901 | 0.907 |
| THieF_1wk-4root        | 3          | 1395.9 | 2278.9 | 0.246  | 0.842  | 0.913 | 0.937 |
| THieF_2wk-noTransform  | 3          | 1493.7 | 2269.1 | 0.364  | 0.809  | 0.976 | 0.933 |
| THieF_3wk-noTransform  | 3          | 1511.2 | 2225.6 | 0.371  | 0.743  | 0.988 | 0.915 |
| COVIDhub-baseline      | 3          | 1529.7 | 2433.1 | 0.581  | 0.978  | 1.000 | 1.000 |
| THieF_4wk-4root        | 3          | 1535.8 | 2295.7 | 0.268  | 0.673  | 1.004 | 0.944 |
| THieF_4wk-noTransform  | 3          | 1537.1 | 2272.6 | 0.357  | 0.750  | 1.005 | 0.934 |
| THieF_3wk-4root        | 3          | 1549.1 | 2437.4 | 0.261  | 0.787  | 1.013 | 1.002 |
| THieF_12wk-noTransform | 3          | 1627.6 | 2518.8 | 0.342  | 0.798  | 1.064 | 1.035 |
| sarima_s7-noTransform  | 3          | 1679.5 | 2542.7 | 0.375  | 0.820  | 1.098 | 1.045 |
| sarima_s1-noTransform  | 3          | 1696.4 | 2507.9 | 0.301  | 0.743  | 1.109 | 1.031 |
| sarima_s1-4root        | 3          | 1777.2 | 2573.7 | 0.206  | 0.676  | 1.162 | 1.058 |
| sarima_s7-4root        | 3          | 1818.3 | 2837.9 | 0.195  | 0.776  | 1.189 | 1.166 |
| THieF_12wk-4root       | 3          | 2209.7 | 2155.1 | 0.324  | 0.662  | 1.445 | 0.886 |
| THieF_6wk-noTransform  | 4          | 1614.6 | 2551.0 | 0.317  | 0.883  | 0.841 | 0.829 |
| THieF_8wk-noTransform  | 4          | 1740.6 | 2664.2 | 0.370  | 0.838  | 0.907 | 0.866 |
| THieF_6wk-4root        | 4          | 1744.2 | 2621.4 | 0.249  | 0.702  | 0.909 | 0.852 |
| THieF_8wk-4root        | 4          | 1766.0 | 2616.8 | 0.257  | 0.709  | 0.920 | 0.851 |
| THieF_1wk-noTransform  | 4          | 1871.0 | 3000.8 | 0.468  | 0.891  | 0.975 | 0.975 |
| COVIDhub-baseline      | 4          | 1919.5 | 3076.4 | 0.529  | 0.967  | 1.000 | 1.000 |
| THieF_4wk-noTransform  | 4          | 1954.8 | 2916.1 | 0.272  | 0.732  | 1.018 | 0.948 |
| THieF_2wk-4root        | 4          | 1978.8 | 3114.6 | 0.158  | 0.777  | 1.031 | 1.012 |
| THieF_12wk-noTransform | 4          | 1991.9 | 3043.7 | 0.366  | 0.751  | 1.038 | 0.989 |
| THieF_2wk-noTransform  | 4          | 1998.9 | 3048.8 | 0.343  | 0.792  | 1.041 | 0.991 |
| THieF_4wk-4root        | 4          | 2018.1 | 3002.0 | 0.181  | 0.623  | 1.051 | 0.976 |
| THieF_3wk-noTransform  | 4          | 2038.4 | 2993.0 | 0.343  | 0.766  | 1.062 | 0.973 |
| THieF_1wk-4root        | 4          | 2051.1 | 3316.3 | 0.249  | 0.842  | 1.069 | 1.078 |
| THieF_3wk-4root        | 4          | 2118.0 | 3326.7 | 0.249  | 0.830  | 1.103 | 1.081 |
| sarima_s1-noTransform  | 4          | 2170.3 | 3141.6 | 0.291  | 0.706  | 1.131 | 1.021 |
| sarima_s1-4root        | 4          | 2284.3 | 3218.7 | 0.211  | 0.672  | 1.190 | 1.046 |
| sarima_s7-noTransform  | 4          | 2340.3 | 3568.8 | 0.358  | 0.808  | 1.219 | 1.160 |
| sarima_s7-4root        | 4          | 2697.4 | 4189.4 | 0.181  | 0.736  | 1.405 | 1.362 |
| THieF_12wk-4root       | 4          | 4832.4 | 2796.2 | 0.332  | 0.581  | 2.518 | 0.909 |

| model                 | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-----------------------|------------|--------|--------|--------|--------|-------|-------|
| THieF_2wk-4root       | 1          | 17.959 | 27.779 | 0.523  | 0.937  | 0.805 | 0.929 |
| THieF_1wk-4root       | 1          | 18.266 | 27.836 | 0.524  | 0.933  | 0.819 | 0.931 |
| THieF_2wk-noTransform | 1          | 19.237 | 27.333 | 0.601  | 0.916  | 0.862 | 0.914 |
| THieF_1wk-noTransform | 1          | 19.268 | 27.524 | 0.598  | 0.912  | 0.864 | 0.920 |
| sarima_s7-4root       | 1          | 20.022 | 30.107 | 0.505  | 0.920  | 0.897 | 1.006 |
| sarima_s1-4root       | 1          | 20.133 | 30.394 | 0.505  | 0.924  | 0.902 | 1.016 |
| sarima_s7-noTransform | 1          | 20.699 | 29.235 | 0.583  | 0.902  | 0.928 | 0.977 |

| model                  | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|------------|--------|--------|--------|--------|-------|-------|
| THieF_3wk-4root        | 1          | 20.990 | 31.924 | 0.489  | 0.922  | 0.941 | 1.067 |
| THieF_3wk-noTransform  | 1          | 21.647 | 31.781 | 0.551  | 0.906  | 0.970 | 1.062 |
| THieF_6wk-noTransform  | 1          | 22.217 | 32.910 | 0.521  | 0.918  | 0.996 | 1.100 |
| COVIDhub-baseline      | 1          | 22.310 | 29.915 | 0.816  | 0.981  | 1.000 | 1.000 |
| THieF_4wk-4root        | 1          | 22.399 | 33.832 | 0.467  | 0.903  | 1.004 | 1.131 |
| sarima_s1-noTransform  | 1          | 22.839 | 32.331 | 0.577  | 0.903  | 1.024 | 1.081 |
| THieF_6wk-4root        | 1          | 23.034 | 34.513 | 0.447  | 0.898  | 1.032 | 1.154 |
| THieF_4wk-noTransform  | 1          | 24.362 | 35.779 | 0.514  | 0.896  | 1.092 | 1.196 |
| THieF_8wk-noTransform  | 1          | 26.941 | 39.108 | 0.485  | 0.887  | 1.208 | 1.307 |
| THieF_8wk-4root        | 1          | 27.049 | 39.673 | 0.409  | 0.851  | 1.212 | 1.326 |
| THieF_12wk-4root       | 1          | 31.284 | 44.609 | 0.384  | 0.822  | 1.402 | 1.491 |
| THieF_12wk-noTransform | 1          | 32.063 | 46.715 | 0.387  | 0.854  | 1.437 | 1.562 |
| THieF_1wk-4root        | 2          | 27.118 | 42.031 | 0.491  | 0.924  | 0.851 | 0.972 |
| THieF_6wk-4root        | 2          | 27.148 | 41.687 | 0.468  | 0.921  | 0.852 | 0.964 |
| THieF_2wk-4root        | 2          | 27.297 | 42.323 | 0.479  | 0.919  | 0.856 | 0.979 |
| THieF_6wk-noTransform  | 2          | 27.969 | 42.298 | 0.553  | 0.926  | 0.877 | 0.978 |
| THieF_2wk-noTransform  | 2          | 28.360 | 40.791 | 0.585  | 0.902  | 0.890 | 0.944 |
| THieF_3wk-4root        | 2          | 28.813 | 44.507 | 0.472  | 0.913  | 0.904 | 1.029 |
| THieF_1wk-noTransform  | 2          | 28.815 | 41.786 | 0.586  | 0.903  | 0.904 | 0.967 |
| THieF_4wk-4root        | 2          | 29.114 | 44.423 | 0.465  | 0.906  | 0.913 | 1.028 |
| THieF_3wk-noTransform  | 2          | 29.556 | 43.815 | 0.555  | 0.901  | 0.927 | 1.013 |
| THieF_8wk-4root        | 2          | 30.200 | 45.944 | 0.432  | 0.896  | 0.947 | 1.063 |
| sarima_s7-4root        | 2          | 30.209 | 45.717 | 0.474  | 0.907  | 0.948 | 1.057 |
| sarima_s1-4root        | 2          | 30.551 | 46.367 | 0.465  | 0.901  | 0.958 | 1.072 |
| sarima_s7-noTransform  | 2          | 30.991 | 44.087 | 0.576  | 0.890  | 0.972 | 1.020 |
| THieF_4wk-noTransform  | 2          | 31.021 | 46.196 | 0.536  | 0.899  | 0.973 | 1.069 |
| THieF_8wk-noTransform  | 2          | 31.026 | 46.430 | 0.514  | 0.907  | 0.973 | 1.074 |
| COVIDhub-baseline      | 2          | 31.875 | 43.233 | 0.839  | 0.986  | 1.000 | 1.000 |
| sarima_s1-noTransform  | 2          | 33.894 | 48.541 | 0.559  | 0.884  | 1.063 | 1.123 |
| THieF_12wk-noTransform | 2          | 35.126 | 53.252 | 0.430  | 0.900  | 1.102 | 1.232 |
| THieF_12wk-4root       | 2          | 37.923 | 49.396 | 0.417  | 0.873  | 1.190 | 1.143 |
| THieF_6wk-4root        | 3          | 34.004 | 52.027 | 0.465  | 0.916  | 0.836 | 0.913 |
| THieF_8wk-4root        | 3          | 35.416 | 54.220 | 0.437  | 0.905  | 0.870 | 0.952 |
| THieF_6wk-noTransform  | 3          | 35.483 | 53.902 | 0.547  | 0.918  | 0.872 | 0.946 |
| THieF_8wk-noTransform  | 3          | 37.044 | 55.968 | 0.524  | 0.906  | 0.910 | 0.983 |
| THieF_1wk-4root        | 3          | 37.284 | 57.435 | 0.463  | 0.905  | 0.916 | 1.008 |
| THieF_2wk-4root        | 3          | 37.587 | 57.458 | 0.460  | 0.902  | 0.924 | 1.009 |
| THieF_4wk-4root        | 3          | 37.775 | 57.268 | 0.454  | 0.896  | 0.928 | 1.005 |
| THieF_2wk-noTransform  | 3          | 37.887 | 55.006 | 0.566  | 0.895  | 0.931 | 0.966 |
| THieF_1wk-noTransform  | 3          | 38.576 | 56.482 | 0.567  | 0.894  | 0.948 | 0.992 |
| THieF_3wk-4root        | 3          | 39.049 | 59.719 | 0.448  | 0.888  | 0.960 | 1.048 |
| THieF_4wk-noTransform  | 3          | 39.432 | 58.444 | 0.540  | 0.889  | 0.969 | 1.026 |
| THieF_3wk-noTransform  | 3          | 39.569 | 58.925 | 0.527  | 0.891  | 0.972 | 1.035 |
| COVIDhub-baseline      | 3          | 40.692 | 56.960 | 0.831  | 0.983  | 1.000 | 1.000 |
| THieF_12wk-noTransform | 3          | 41.035 | 62.631 | 0.434  | 0.905  | 1.008 | 1.100 |
| sarima_s7-noTransform  | 3          | 41.576 | 59.557 | 0.561  | 0.882  | 1.022 | 1.046 |
| sarima_s7-4root        | 3          | 41.584 | 62.634 | 0.451  | 0.892  | 1.022 | 1.100 |
| sarima_s1-4root        | 3          | 42.305 | 63.714 | 0.438  | 0.875  | 1.040 | 1.119 |
| sarima_s1-noTransform  | 3          | 45.135 | 64.745 | 0.530  | 0.868  | 1.109 | 1.137 |
| THieF_12wk-4root       | 3          | 71.204 | 56.917 | 0.419  | 0.884  | 1.750 | 0.999 |
| THieF_6wk-4root        | 4          | 42.426 | 64.206 | 0.445  | 0.897  | 0.872 | 0.920 |
| THieF_8wk-4root        | 4          | 42.713 | 64.729 | 0.433  | 0.897  | 0.878 | 0.927 |



| model                  | horizon_wk | wis     | mae    | cov_50 | cov_95 | rwis  | rmae  |
|------------------------|------------|---------|--------|--------|--------|-------|-------|
| THieF_6wk-noTransform  | 4          | 43.355  | 65.585 | 0.537  | 0.907  | 0.891 | 0.940 |
| THieF_8wk-noTransform  | 4          | 44.034  | 66.723 | 0.527  | 0.900  | 0.905 | 0.956 |
| THieF_4wk-4root        | 4          | 47.095  | 70.483 | 0.436  | 0.879  | 0.968 | 1.010 |
| THieF_2wk-noTransform  | 4          | 47.550  | 68.979 | 0.543  | 0.886  | 0.978 | 0.988 |
| THieF_4wk-noTransform  | 4          | 47.717  | 70.859 | 0.527  | 0.881  | 0.981 | 1.015 |
| THieF_1wk-4root        | 4          | 48.046  | 72.948 | 0.444  | 0.881  | 0.988 | 1.045 |
| THieF_1wk-noTransform  | 4          | 48.284  | 71.043 | 0.552  | 0.883  | 0.993 | 1.018 |
| THieF_12wk-noTransform | 4          | 48.336  | 73.630 | 0.430  | 0.903  | 0.994 | 1.055 |
| COVIDhub-baseline      | 4          | 48.633  | 69.792 | 0.820  | 0.982  | 1.000 | 1.000 |
| THieF_2wk-4root        | 4          | 48.899  | 73.473 | 0.435  | 0.877  | 1.005 | 1.053 |
| THieF_3wk-noTransform  | 4          | 49.069  | 73.786 | 0.507  | 0.887  | 1.009 | 1.057 |
| THieF_3wk-4root        | 4          | 49.637  | 74.682 | 0.432  | 0.873  | 1.021 | 1.070 |
| sarima_s7-noTransform  | 4          | 51.901  | 74.747 | 0.546  | 0.870  | 1.067 | 1.071 |
| sarima_s7-4root        | 4          | 53.568  | 79.738 | 0.432  | 0.874  | 1.101 | 1.143 |
| sarima_s1-4root        | 4          | 54.665  | 81.245 | 0.413  | 0.848  | 1.124 | 1.164 |
| sarima_s1-noTransform  | 4          | 55.781  | 79.897 | 0.506  | 0.854  | 1.147 | 1.145 |
| THieF_12wk-4root       | 4          | 196.873 | 67.049 | 0.410  | 0.877  | 4.048 | 0.961 |



When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8\_arima\_noTrans model, which showed more comparable performance to the baseline. The Base\_arima\_noTrans model performed the best in terms of PI Coverage, followed by the Base\_arima\_4root and Topmost8\_arima\_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base\_arima\_4root model performed the best, followed by the Base\_arima\_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunk WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base\_arima\_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8\_arima\_noTrans and Base\_arima\_noTrans models, then the Topmost8\_arima\_4root and Multiple3\_arima\_noTrans models, all of which are much better than the Multiple3\_arima\_4root model.

### Metrics by wave

| wave  | model                 | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|-----------------------|------------|--------|--------|--------|--------|-------|-------|
| alpha | THieF_1wk-4root       | 1          | 294.8  | 471.1  | 0.348  | 0.902  | 0.592 | 0.866 |
| alpha | THieF_2wk-4root       | 1          | 303.5  | 486.9  | 0.357  | 0.911  | 0.609 | 0.895 |
| alpha | THieF_1wk-noTransform | 1          | 316.6  | 495.4  | 0.571  | 0.982  | 0.636 | 0.910 |
| alpha | sarima_s1-4root       | 1          | 321.3  | 521.9  | 0.473  | 0.929  | 0.645 | 0.959 |
| alpha | THieF_2wk-noTransform | 1          | 326.4  | 493.2  | 0.661  | 1.000  | 0.655 | 0.906 |
| alpha | sarima_s7-noTransform | 1          | 353.5  | 568.5  | 0.518  | 0.964  | 0.710 | 1.044 |
| alpha | sarima_s1-noTransform | 1          | 363.5  | 526.0  | 0.786  | 1.000  | 0.730 | 0.966 |
| alpha | sarima_s7-4root       | 1          | 392.2  | 611.4  | 0.196  | 0.750  | 0.787 | 1.123 |
| alpha | THieF_3wk-noTransform | 1          | 457.0  | 750.2  | 0.438  | 0.955  | 0.917 | 1.378 |
| alpha | THieF_3wk-4root       | 1          | 457.5  | 719.8  | 0.188  | 0.768  | 0.918 | 1.323 |
| alpha | THieF_4wk-4root       | 1          | 485.4  | 750.2  | 0.277  | 0.750  | 0.974 | 1.378 |
| alpha | COVIDhub-baseline     | 1          | 498.2  | 544.2  | 0.973  | 1.000  | 1.000 | 1.000 |
| alpha | THieF_4wk-noTransform | 1          | 523.1  | 871.0  | 0.357  | 0.955  | 1.050 | 1.600 |
| alpha | THieF_8wk-noTransform | 1          | 545.1  | 923.2  | 0.393  | 1.000  | 1.094 | 1.696 |
| alpha | THieF_6wk-4root       | 1          | 560.5  | 839.3  | 0.268  | 0.545  | 1.125 | 1.542 |
| alpha | THieF_6wk-noTransform | 1          | 627.6  | 1062.1 | 0.259  | 0.982  | 1.260 | 1.952 |
| alpha | THieF_8wk-4root       | 1          | 728.6  | 1064.3 | 0.143  | 0.518  | 1.462 | 1.955 |
| alpha | THieF_12wk-4root      | 1          | 1341.1 | 1775.3 | 0.134  | 0.455  | 2.692 | 3.262 |

| wave  | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| alpha | THieF_12wk-<br>noTransform | 1          | 1584.5 | 2371.6 | 0.071  | 0.571  | 3.181 | 4.358 |
| alpha | THieF_1wk-4root            | 2          | 519.7  | 834.7  | 0.339  | 0.893  | 0.704 | 0.927 |
| alpha | THieF_2wk-<br>noTransform  | 2          | 525.4  | 831.9  | 0.625  | 1.000  | 0.711 | 0.924 |
| alpha | THieF_2wk-4root            | 2          | 530.8  | 858.4  | 0.321  | 0.911  | 0.719 | 0.953 |
| alpha | sarima_s1-4root            | 2          | 553.1  | 890.2  | 0.286  | 0.866  | 0.749 | 0.989 |
| alpha | sarima_s1-<br>noTransform  | 2          | 554.3  | 885.6  | 0.616  | 1.000  | 0.750 | 0.984 |
| alpha | THieF_1wk-<br>noTransform  | 2          | 575.8  | 927.0  | 0.527  | 1.000  | 0.779 | 1.030 |
| alpha | sarima_s7-<br>noTransform  | 2          | 623.3  | 1003.9 | 0.598  | 0.982  | 0.844 | 1.115 |
| alpha | THieF_3wk-<br>noTransform  | 2          | 652.4  | 1060.2 | 0.473  | 1.000  | 0.883 | 1.178 |
| alpha | THieF_4wk-4root            | 2          | 653.3  | 1005.1 | 0.304  | 0.777  | 0.884 | 1.116 |
| alpha | sarima_s7-4root            | 2          | 659.6  | 1040.2 | 0.196  | 0.830  | 0.893 | 1.155 |
| alpha | THieF_8wk-<br>noTransform  | 2          | 670.7  | 1103.6 | 0.616  | 1.000  | 0.908 | 1.226 |
| alpha | THieF_3wk-4root            | 2          | 675.1  | 1058.9 | 0.295  | 0.777  | 0.914 | 1.176 |
| alpha | THieF_4wk-<br>noTransform  | 2          | 694.8  | 1149.7 | 0.402  | 1.000  | 0.941 | 1.277 |
| alpha | THieF_6wk-4root            | 2          | 702.7  | 1080.0 | 0.250  | 0.705  | 0.951 | 1.200 |
| alpha | COVIDhub-baseline          | 2          | 738.7  | 900.3  | 0.982  | 1.000  | 1.000 | 1.000 |
| alpha | THieF_6wk-<br>noTransform  | 2          | 773.2  | 1298.1 | 0.375  | 1.000  | 1.047 | 1.442 |
| alpha | THieF_8wk-4root            | 2          | 810.3  | 1224.9 | 0.277  | 0.616  | 1.097 | 1.361 |
| alpha | THieF_12wk-<br>noTransform | 2          | 1561.3 | 2532.8 | 0.098  | 0.741  | 2.114 | 2.813 |
| alpha | THieF_12wk-4root           | 2          | 1630.7 | 1791.7 | 0.268  | 0.554  | 2.207 | 1.990 |
| alpha | THieF_2wk-<br>noTransform  | 3          | 798.5  | 1303.2 | 0.554  | 1.000  | 0.818 | 0.979 |
| alpha | sarima_s1-<br>noTransform  | 3          | 804.1  | 1312.1 | 0.473  | 0.973  | 0.824 | 0.986 |
| alpha | THieF_1wk-4root            | 3          | 840.8  | 1347.1 | 0.241  | 0.830  | 0.862 | 1.012 |
| alpha | THieF_2wk-4root            | 3          | 859.6  | 1378.4 | 0.188  | 0.839  | 0.881 | 1.035 |
| alpha | sarima_s1-4root            | 3          | 874.6  | 1320.0 | 0.250  | 0.750  | 0.896 | 0.991 |
| alpha | THieF_1wk-<br>noTransform  | 3          | 906.2  | 1511.5 | 0.562  | 1.000  | 0.929 | 1.135 |
| alpha | THieF_8wk-<br>noTransform  | 3          | 909.1  | 1469.1 | 0.491  | 1.000  | 0.932 | 1.103 |
| alpha | THieF_4wk-<br>noTransform  | 3          | 935.2  | 1556.1 | 0.366  | 0.991  | 0.958 | 1.169 |
| alpha | THieF_4wk-4root            | 3          | 935.4  | 1412.8 | 0.295  | 0.688  | 0.959 | 1.061 |
| alpha | sarima_s7-<br>noTransform  | 3          | 951.7  | 1517.6 | 0.562  | 1.000  | 0.975 | 1.140 |
| alpha | COVIDhub-baseline          | 3          | 975.7  | 1331.3 | 0.920  | 1.000  | 1.000 | 1.000 |
| alpha | THieF_3wk-<br>noTransform  | 3          | 999.0  | 1659.5 | 0.384  | 0.955  | 1.024 | 1.246 |
| alpha | THieF_6wk-<br>noTransform  | 3          | 1004.2 | 1716.1 | 0.348  | 1.000  | 1.029 | 1.289 |
| alpha | THieF_6wk-4root            | 3          | 1012.9 | 1552.3 | 0.205  | 0.634  | 1.038 | 1.166 |

| wave  | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| alpha | sarima_s7-4root            | 3          | 1013.1 | 1587.5 | 0.214  | 0.804  | 1.038 | 1.192 |
| alpha | THieF_8wk-4root            | 3          | 1053.6 | 1610.9 | 0.214  | 0.607  | 1.080 | 1.210 |
| alpha | THieF_3wk-4root            | 3          | 1076.9 | 1669.5 | 0.188  | 0.768  | 1.104 | 1.254 |
| alpha | THieF_12wk-<br>noTransform | 3          | 1769.7 | 2885.7 | 0.179  | 0.812  | 1.814 | 2.168 |
| alpha | THieF_12wk-4root           | 3          | 3325.1 | 2161.3 | 0.179  | 0.491  | 3.408 | 1.623 |
| alpha | THieF_2wk-<br>noTransform  | 4          | 1107.7 | 1838.2 | 0.536  | 1.000  | 0.877 | 0.993 |
| alpha | sarima_s1-<br>noTransform  | 4          | 1150.3 | 1830.6 | 0.420  | 0.929  | 0.911 | 0.989 |
| alpha | THieF_8wk-<br>noTransform  | 4          | 1202.6 | 1900.7 | 0.438  | 0.973  | 0.952 | 1.027 |
| alpha | THieF_6wk-<br>noTransform  | 4          | 1262.4 | 2147.4 | 0.312  | 0.964  | 0.999 | 1.160 |
| alpha | COVIDhub-baseline          | 4          | 1263.2 | 1851.0 | 0.857  | 1.000  | 1.000 | 1.000 |
| alpha | THieF_2wk-4root            | 4          | 1264.0 | 1996.3 | 0.170  | 0.759  | 1.001 | 1.079 |
| alpha | THieF_1wk-4root            | 4          | 1275.7 | 2009.1 | 0.232  | 0.812  | 1.010 | 1.085 |
| alpha | sarima_s1-4root            | 4          | 1289.7 | 1840.6 | 0.223  | 0.723  | 1.021 | 0.994 |
| alpha | THieF_1wk-<br>noTransform  | 4          | 1318.0 | 2254.6 | 0.518  | 1.000  | 1.043 | 1.218 |
| alpha | THieF_4wk-<br>noTransform  | 4          | 1343.4 | 2220.5 | 0.268  | 0.929  | 1.064 | 1.200 |
| alpha | sarima_s7-<br>noTransform  | 4          | 1360.3 | 2189.6 | 0.527  | 1.000  | 1.077 | 1.183 |
| alpha | THieF_4wk-4root            | 4          | 1402.1 | 2048.1 | 0.170  | 0.616  | 1.110 | 1.106 |
| alpha | THieF_8wk-4root            | 4          | 1429.3 | 2129.4 | 0.188  | 0.652  | 1.132 | 1.150 |
| alpha | THieF_6wk-4root            | 4          | 1461.7 | 2138.9 | 0.161  | 0.598  | 1.157 | 1.156 |
| alpha | sarima_s7-4root            | 4          | 1471.2 | 2307.3 | 0.188  | 0.759  | 1.165 | 1.246 |
| alpha | THieF_3wk-<br>noTransform  | 4          | 1486.9 | 2464.2 | 0.295  | 0.938  | 1.177 | 1.331 |
| alpha | THieF_3wk-4root            | 4          | 1638.3 | 2543.9 | 0.170  | 0.795  | 1.297 | 1.374 |
| alpha | THieF_12wk-<br>noTransform | 4          | 2139.2 | 3471.8 | 0.179  | 0.768  | 1.694 | 1.876 |
| alpha | THieF_12wk-4root           | 4          | 8990.8 | 2925.8 | 0.179  | 0.375  | 7.118 | 1.581 |
| delta | THieF_2wk-4root            | 1          | 553.9  | 947.1  | 0.234  | 0.973  | 0.720 | 0.722 |
| delta | THieF_1wk-4root            | 1          | 578.8  | 951.2  | 0.216  | 0.892  | 0.753 | 0.725 |
| delta | THieF_1wk-<br>noTransform  | 1          | 600.5  | 951.7  | 0.216  | 0.802  | 0.781 | 0.726 |
| delta | THieF_2wk-<br>noTransform  | 1          | 616.0  | 1004.2 | 0.288  | 0.874  | 0.801 | 0.766 |
| delta | THieF_3wk-4root            | 1          | 677.7  | 1081.3 | 0.270  | 0.829  | 0.881 | 0.825 |
| delta | THieF_6wk-4root            | 1          | 685.5  | 1112.9 | 0.360  | 0.928  | 0.892 | 0.849 |
| delta | THieF_3wk-<br>noTransform  | 1          | 740.8  | 1151.4 | 0.315  | 0.820  | 0.963 | 0.878 |
| delta | THieF_6wk-<br>noTransform  | 1          | 743.7  | 1176.4 | 0.288  | 0.856  | 0.967 | 0.897 |
| delta | THieF_12wk-4root           | 1          | 752.8  | 1115.1 | 0.523  | 0.865  | 0.979 | 0.850 |
| delta | THieF_4wk-<br>noTransform  | 1          | 764.3  | 1181.6 | 0.324  | 0.784  | 0.994 | 0.901 |
| delta | COVIDhub-baseline          | 1          | 768.8  | 1311.2 | 0.450  | 1.000  | 1.000 | 1.000 |
| delta | sarima_s7-4root            | 1          | 786.2  | 1237.0 | 0.144  | 0.730  | 1.023 | 0.943 |

| wave  | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| delta | sarima_s1-<br>noTransform  | 1          | 822.7  | 1336.0 | 0.297  | 0.874  | 1.070 | 1.019 |
| delta | sarima_s1-4root            | 1          | 825.3  | 1310.6 | 0.315  | 0.892  | 1.073 | 1.000 |
| delta | sarima_s7-<br>noTransform  | 1          | 825.8  | 1226.2 | 0.144  | 0.631  | 1.074 | 0.935 |
| delta | THieF_4wk-4root            | 1          | 859.1  | 1341.9 | 0.306  | 0.793  | 1.117 | 1.023 |
| delta | THieF_8wk-<br>noTransform  | 1          | 873.0  | 1381.6 | 0.306  | 0.829  | 1.135 | 1.054 |
| delta | THieF_12wk-<br>noTransform | 1          | 1045.7 | 1482.8 | 0.450  | 0.757  | 1.360 | 1.131 |
| delta | THieF_8wk-4root            | 1          | 1083.0 | 1658.5 | 0.216  | 0.721  | 1.409 | 1.265 |
| delta | THieF_6wk-4root            | 2          | 996.3  | 1529.2 | 0.471  | 0.837  | 0.719 | 0.643 |
| delta | THieF_6wk-<br>noTransform  | 2          | 1048.0 | 1715.9 | 0.240  | 0.904  | 0.756 | 0.722 |
| delta | THieF_12wk-4root           | 2          | 1060.4 | 1538.0 | 0.577  | 0.846  | 0.765 | 0.647 |
| delta | THieF_2wk-4root            | 2          | 1149.1 | 1872.5 | 0.183  | 0.846  | 0.829 | 0.788 |
| delta | THieF_1wk-<br>noTransform  | 2          | 1173.4 | 1881.3 | 0.144  | 0.760  | 0.847 | 0.791 |
| delta | THieF_8wk-<br>noTransform  | 2          | 1193.1 | 1950.2 | 0.269  | 0.865  | 0.861 | 0.820 |
| delta | THieF_1wk-4root            | 2          | 1197.0 | 1957.6 | 0.192  | 0.837  | 0.864 | 0.823 |
| delta | THieF_2wk-<br>noTransform  | 2          | 1252.1 | 1928.3 | 0.183  | 0.683  | 0.904 | 0.811 |
| delta | THieF_3wk-4root            | 2          | 1273.5 | 2066.2 | 0.221  | 0.865  | 0.919 | 0.869 |
| delta | THieF_3wk-<br>noTransform  | 2          | 1279.3 | 1906.4 | 0.308  | 0.615  | 0.923 | 0.802 |
| delta | THieF_4wk-<br>noTransform  | 2          | 1316.6 | 2010.3 | 0.269  | 0.663  | 0.950 | 0.846 |
| delta | THieF_12wk-<br>noTransform | 2          | 1343.6 | 1925.9 | 0.481  | 0.769  | 0.970 | 0.810 |
| delta | THieF_8wk-4root            | 2          | 1357.0 | 2057.2 | 0.288  | 0.760  | 0.979 | 0.865 |
| delta | COVIDhub-baseline          | 2          | 1385.6 | 2377.6 | 0.327  | 0.990  | 1.000 | 1.000 |
| delta | THieF_4wk-4root            | 2          | 1469.5 | 2255.9 | 0.279  | 0.702  | 1.061 | 0.949 |
| delta | sarima_s1-<br>noTransform  | 2          | 1581.6 | 2425.5 | 0.144  | 0.712  | 1.141 | 1.020 |
| delta | sarima_s1-4root            | 2          | 1597.4 | 2397.0 | 0.192  | 0.731  | 1.153 | 1.008 |
| delta | sarima_s7-<br>noTransform  | 2          | 1612.9 | 2428.3 | 0.038  | 0.625  | 1.164 | 1.021 |
| delta | sarima_s7-4root            | 2          | 1625.7 | 2554.2 | 0.115  | 0.712  | 1.173 | 1.074 |
| delta | THieF_12wk-4root           | 3          | 1322.7 | 1931.5 | 0.577  | 0.804  | 0.657 | 0.566 |
| delta | THieF_6wk-4root            | 3          | 1412.2 | 2142.1 | 0.526  | 0.845  | 0.701 | 0.628 |
| delta | THieF_6wk-<br>noTransform  | 3          | 1492.5 | 2367.4 | 0.309  | 0.825  | 0.741 | 0.694 |
| delta | THieF_8wk-<br>noTransform  | 3          | 1519.0 | 2453.4 | 0.340  | 0.866  | 0.754 | 0.719 |
| delta | THieF_12wk-<br>noTransform | 3          | 1620.3 | 2295.2 | 0.536  | 0.722  | 0.804 | 0.673 |
| delta | THieF_8wk-4root            | 3          | 1621.4 | 2391.5 | 0.330  | 0.794  | 0.805 | 0.701 |
| delta | THieF_2wk-4root            | 3          | 1865.2 | 2962.6 | 0.227  | 0.825  | 0.926 | 0.868 |
| delta | THieF_1wk-<br>noTransform  | 3          | 1869.2 | 2988.8 | 0.186  | 0.722  | 0.928 | 0.876 |

| wave     | model                  | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|----------|------------------------|------------|--------|--------|--------|--------|-------|-------|
| delta    | THieF_4wk-noTransform  | 3          | 1909.4 | 2806.1 | 0.247  | 0.526  | 0.948 | 0.823 |
| delta    | THieF_2wk-noTransform  | 3          | 1921.1 | 2854.6 | 0.258  | 0.608  | 0.954 | 0.837 |
| delta    | THieF_3wk-noTransform  | 3          | 1928.9 | 2777.7 | 0.227  | 0.474  | 0.958 | 0.814 |
| delta    | THieF_3wk-4root        | 3          | 1981.2 | 3229.3 | 0.361  | 0.897  | 0.984 | 0.947 |
| delta    | COVIDhub-baseline      | 3          | 2014.3 | 3411.3 | 0.216  | 0.938  | 1.000 | 1.000 |
| delta    | THieF_1wk-4root        | 3          | 2073.5 | 3422.0 | 0.196  | 0.856  | 1.029 | 1.003 |
| delta    | THieF_4wk-4root        | 3          | 2081.4 | 3156.4 | 0.247  | 0.680  | 1.033 | 0.925 |
| delta    | sarima_s1-noTransform  | 3          | 2364.2 | 3459.1 | 0.103  | 0.577  | 1.174 | 1.014 |
| delta    | sarima_s1-4root        | 3          | 2370.7 | 3429.1 | 0.144  | 0.691  | 1.177 | 1.005 |
| delta    | sarima_s7-noTransform  | 3          | 2603.9 | 3891.1 | 0.062  | 0.567  | 1.293 | 1.141 |
| delta    | sarima_s7-4root        | 3          | 2900.4 | 4517.3 | 0.113  | 0.742  | 1.440 | 1.324 |
| delta    | THieF_12wk-4root       | 4          | 1667.0 | 2527.1 | 0.533  | 0.844  | 0.647 | 0.584 |
| delta    | THieF_12wk-noTransform | 4          | 1954.4 | 2784.1 | 0.522  | 0.722  | 0.759 | 0.644 |
| delta    | THieF_6wk-4root        | 4          | 1986.2 | 3007.4 | 0.344  | 0.844  | 0.771 | 0.696 |
| delta    | THieF_8wk-noTransform  | 4          | 2012.7 | 3190.0 | 0.278  | 0.811  | 0.782 | 0.738 |
| delta    | THieF_6wk-noTransform  | 4          | 2058.4 | 3152.0 | 0.211  | 0.778  | 0.799 | 0.729 |
| delta    | THieF_8wk-4root        | 4          | 2079.2 | 3067.3 | 0.278  | 0.800  | 0.807 | 0.709 |
| delta    | THieF_4wk-noTransform  | 4          | 2443.0 | 3555.4 | 0.200  | 0.533  | 0.949 | 0.822 |
| delta    | THieF_3wk-noTransform  | 4          | 2562.4 | 3714.4 | 0.189  | 0.544  | 0.995 | 0.859 |
| delta    | COVIDhub-baseline      | 4          | 2574.9 | 4323.7 | 0.156  | 0.900  | 1.000 | 1.000 |
| delta    | THieF_1wk-noTransform  | 4          | 2638.8 | 4183.7 | 0.278  | 0.756  | 1.025 | 0.968 |
| delta    | THieF_3wk-4root        | 4          | 2688.3 | 4407.1 | 0.333  | 0.967  | 1.044 | 1.019 |
| delta    | THieF_2wk-noTransform  | 4          | 2716.1 | 3950.3 | 0.211  | 0.533  | 1.055 | 0.914 |
| delta    | THieF_4wk-4root        | 4          | 2739.3 | 4173.8 | 0.189  | 0.622  | 1.064 | 0.965 |
| delta    | THieF_2wk-4root        | 4          | 2879.3 | 4516.4 | 0.167  | 0.722  | 1.118 | 1.045 |
| delta    | sarima_s1-4root        | 4          | 3051.6 | 4343.4 | 0.133  | 0.711  | 1.185 | 1.005 |
| delta    | sarima_s1-noTransform  | 4          | 3059.1 | 4374.8 | 0.089  | 0.522  | 1.188 | 1.012 |
| delta    | THieF_1wk-4root        | 4          | 3232.4 | 5301.8 | 0.278  | 0.844  | 1.255 | 1.226 |
| delta    | sarima_s7-noTransform  | 4          | 3746.3 | 5558.1 | 0.089  | 0.511  | 1.455 | 1.285 |
| delta    | sarima_s7-4root        | 4          | 4703.2 | 7253.4 | 0.156  | 0.678  | 1.827 | 1.678 |
| winter21 | THieF_6wk-noTransform  | 1          | 480.4  | 792.8  | 0.476  | 1.000  | 0.592 | 0.632 |
| winter21 | THieF_1wk-noTransform  | 1          | 505.9  | 681.6  | 0.714  | 0.889  | 0.624 | 0.543 |
| winter21 | THieF_1wk-4root        | 1          | 527.6  | 860.1  | 0.508  | 0.937  | 0.650 | 0.685 |
| winter21 | sarima_s7-noTransform  | 1          | 564.3  | 791.5  | 0.619  | 0.889  | 0.696 | 0.631 |
| winter21 | sarima_s7-4root        | 1          | 635.4  | 1010.9 | 0.476  | 0.921  | 0.783 | 0.806 |

| wave     | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|----------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| winter21 | THieF_2wk-4root            | 1          | 669.6  | 1094.9 | 0.381  | 0.937  | 0.825 | 0.873 |
| winter21 | COVIDhub-baseline          | 1          | 811.3  | 1254.9 | 0.686  | 1.000  | 1.000 | 1.000 |
| winter21 | THieF_6wk-4root            | 1          | 846.9  | 1385.2 | 0.286  | 0.873  | 1.044 | 1.104 |
| winter21 | THieF_2wk-<br>noTransform  | 1          | 910.8  | 1335.9 | 0.381  | 0.778  | 1.123 | 1.065 |
| winter21 | THieF_12wk-<br>noTransform | 1          | 951.5  | 1588.7 | 0.175  | 0.905  | 1.173 | 1.266 |
| winter21 | sarima_s1-<br>noTransform  | 1          | 994.6  | 1534.4 | 0.349  | 0.810  | 1.226 | 1.223 |
| winter21 | THieF_3wk-4root            | 1          | 1010.2 | 1615.0 | 0.143  | 0.778  | 1.245 | 1.287 |
| winter21 | THieF_4wk-4root            | 1          | 1077.8 | 1676.5 | 0.143  | 0.698  | 1.328 | 1.336 |
| winter21 | THieF_8wk-4root            | 1          | 1124.1 | 1789.7 | 0.079  | 0.635  | 1.386 | 1.426 |
| winter21 | THieF_3wk-<br>noTransform  | 1          | 1134.5 | 1631.5 | 0.238  | 0.698  | 1.398 | 1.300 |
| winter21 | sarima_s1-4root            | 1          | 1144.6 | 1775.7 | 0.238  | 0.794  | 1.411 | 1.415 |
| winter21 | THieF_12wk-4root           | 1          | 1251.9 | 1935.7 | 0.127  | 0.571  | 1.543 | 1.543 |
| winter21 | THieF_4wk-<br>noTransform  | 1          | 1261.2 | 1824.1 | 0.254  | 0.651  | 1.555 | 1.454 |
| winter21 | THieF_8wk-<br>noTransform  | 1          | 1362.8 | 2035.3 | 0.143  | 0.508  | 1.680 | 1.622 |
| winter21 | THieF_6wk-<br>noTransform  | 2          | 703.3  | 1064.1 | 0.587  | 0.952  | 0.555 | 0.521 |
| winter21 | THieF_1wk-<br>noTransform  | 2          | 885.4  | 1176.7 | 0.683  | 0.889  | 0.698 | 0.577 |
| winter21 | THieF_1wk-4root            | 2          | 934.8  | 1511.0 | 0.365  | 0.889  | 0.737 | 0.740 |
| winter21 | sarima_s7-<br>noTransform  | 2          | 1008.3 | 1418.3 | 0.635  | 0.889  | 0.795 | 0.695 |
| winter21 | THieF_12wk-<br>noTransform | 2          | 1068.7 | 1835.1 | 0.190  | 0.921  | 0.843 | 0.899 |
| winter21 | THieF_2wk-4root            | 2          | 1078.6 | 1779.2 | 0.222  | 0.889  | 0.851 | 0.872 |
| winter21 | THieF_6wk-4root            | 2          | 1092.9 | 1811.0 | 0.302  | 0.921  | 0.862 | 0.887 |
| winter21 | sarima_s7-4root            | 2          | 1125.7 | 1740.5 | 0.413  | 0.841  | 0.888 | 0.853 |
| winter21 | COVIDhub-baseline          | 2          | 1267.7 | 2040.6 | 0.657  | 1.000  | 1.000 | 1.000 |
| winter21 | THieF_8wk-4root            | 2          | 1315.6 | 2138.9 | 0.127  | 0.825  | 1.038 | 1.048 |
| winter21 | THieF_3wk-4root            | 2          | 1345.3 | 2059.0 | 0.222  | 0.778  | 1.061 | 1.009 |
| winter21 | THieF_12wk-4root           | 2          | 1354.7 | 2202.9 | 0.095  | 0.778  | 1.069 | 1.080 |
| winter21 | THieF_3wk-<br>noTransform  | 2          | 1407.9 | 1946.9 | 0.476  | 0.714  | 1.111 | 0.954 |
| winter21 | THieF_4wk-4root            | 2          | 1431.2 | 2051.2 | 0.429  | 0.667  | 1.129 | 1.005 |
| winter21 | THieF_2wk-<br>noTransform  | 2          | 1454.2 | 2138.1 | 0.270  | 0.794  | 1.147 | 1.048 |
| winter21 | THieF_8wk-<br>noTransform  | 2          | 1599.9 | 2479.8 | 0.190  | 0.714  | 1.262 | 1.215 |
| winter21 | THieF_4wk-<br>noTransform  | 2          | 1683.3 | 2173.6 | 0.492  | 0.651  | 1.328 | 1.065 |
| winter21 | sarima_s1-<br>noTransform  | 2          | 1686.2 | 2466.1 | 0.254  | 0.746  | 1.330 | 1.209 |
| winter21 | sarima_s1-4root            | 2          | 1865.8 | 2747.0 | 0.190  | 0.603  | 1.472 | 1.346 |
| winter21 | THieF_6wk-<br>noTransform  | 3          | 1150.3 | 1763.6 | 0.492  | 0.921  | 0.659 | 0.621 |
| winter21 | THieF_1wk-<br>noTransform  | 3          | 1311.4 | 1821.7 | 0.651  | 0.889  | 0.752 | 0.641 |



| wave     | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|----------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| winter21 | THieF_1wk-4root            | 3          | 1339.3 | 2175.3 | 0.333  | 0.841  | 0.768 | 0.766 |
| winter21 | THieF_12wk-<br>noTransform | 3          | 1386.5 | 2210.8 | 0.333  | 0.889  | 0.795 | 0.778 |
| winter21 | THieF_6wk-4root            | 3          | 1473.5 | 2328.0 | 0.302  | 0.794  | 0.845 | 0.820 |
| winter21 | THieF_2wk-4root            | 3          | 1548.2 | 2518.4 | 0.111  | 0.841  | 0.888 | 0.887 |
| winter21 | sarima_s7-<br>noTransform  | 3          | 1550.1 | 2289.2 | 0.524  | 0.889  | 0.889 | 0.806 |
| winter21 | THieF_8wk-4root            | 3          | 1578.2 | 2410.7 | 0.302  | 0.762  | 0.905 | 0.849 |
| winter21 | sarima_s7-4root            | 3          | 1583.8 | 2475.1 | 0.286  | 0.778  | 0.908 | 0.871 |
| winter21 | THieF_12wk-4root           | 3          | 1592.4 | 2488.5 | 0.190  | 0.746  | 0.913 | 0.876 |
| winter21 | THieF_3wk-4root            | 3          | 1723.0 | 2583.5 | 0.238  | 0.651  | 0.988 | 0.910 |
| winter21 | COVIDhub-baseline          | 3          | 1744.4 | 2840.4 | 0.543  | 1.000  | 1.000 | 1.000 |
| winter21 | THieF_4wk-4root            | 3          | 1763.1 | 2540.0 | 0.254  | 0.635  | 1.011 | 0.894 |
| winter21 | THieF_3wk-<br>noTransform  | 3          | 1778.4 | 2382.0 | 0.571  | 0.778  | 1.020 | 0.839 |
| winter21 | THieF_8wk-<br>noTransform  | 3          | 1891.3 | 2718.5 | 0.429  | 0.667  | 1.084 | 0.957 |
| winter21 | THieF_4wk-<br>noTransform  | 3          | 2033.8 | 2725.0 | 0.508  | 0.667  | 1.166 | 0.959 |
| winter21 | THieF_2wk-<br>noTransform  | 3          | 2071.4 | 3084.8 | 0.190  | 0.778  | 1.187 | 1.086 |
| winter21 | sarima_s1-<br>noTransform  | 3          | 2254.4 | 3169.1 | 0.302  | 0.587  | 1.292 | 1.116 |
| winter21 | sarima_s1-4root            | 3          | 2467.8 | 3485.6 | 0.222  | 0.524  | 1.415 | 1.227 |
| winter21 | THieF_6wk-<br>noTransform  | 4          | 1606.7 | 2409.8 | 0.476  | 0.889  | 0.755 | 0.702 |
| winter21 | THieF_1wk-4root            | 4          | 1742.2 | 2803.8 | 0.238  | 0.889  | 0.819 | 0.817 |
| winter21 | THieF_1wk-<br>noTransform  | 4          | 1757.3 | 2637.7 | 0.651  | 0.889  | 0.826 | 0.768 |
| winter21 | THieF_12wk-<br>noTransform | 4          | 1783.6 | 2653.6 | 0.476  | 0.762  | 0.839 | 0.773 |
| winter21 | THieF_6wk-4root            | 4          | 1900.7 | 2927.7 | 0.270  | 0.683  | 0.894 | 0.853 |
| winter21 | THieF_8wk-4root            | 4          | 1917.2 | 2839.9 | 0.349  | 0.683  | 0.901 | 0.827 |
| winter21 | THieF_12wk-4root           | 4          | 1961.6 | 2950.2 | 0.317  | 0.571  | 0.922 | 0.859 |
| winter21 | THieF_2wk-4root            | 4          | 1963.2 | 3100.0 | 0.127  | 0.889  | 0.923 | 0.903 |
| winter21 | sarima_s7-4root            | 4          | 2011.9 | 3158.3 | 0.206  | 0.778  | 0.946 | 0.920 |
| winter21 | sarima_s7-<br>noTransform  | 4          | 2074.0 | 3178.7 | 0.444  | 0.889  | 0.975 | 0.926 |
| winter21 | THieF_4wk-4root            | 4          | 2082.9 | 3023.7 | 0.190  | 0.635  | 0.979 | 0.881 |
| winter21 | COVIDhub-baseline          | 4          | 2126.8 | 3433.3 | 0.486  | 1.000  | 1.000 | 1.000 |
| winter21 | THieF_3wk-4root            | 4          | 2156.2 | 3174.9 | 0.270  | 0.698  | 1.014 | 0.925 |
| winter21 | THieF_3wk-<br>noTransform  | 4          | 2270.2 | 2902.4 | 0.651  | 0.778  | 1.067 | 0.845 |
| winter21 | THieF_8wk-<br>noTransform  | 4          | 2308.4 | 3270.5 | 0.381  | 0.635  | 1.085 | 0.953 |
| winter21 | THieF_4wk-<br>noTransform  | 4          | 2344.1 | 3239.5 | 0.381  | 0.667  | 1.102 | 0.944 |
| winter21 | THieF_2wk-<br>noTransform  | 4          | 2558.5 | 3913.1 | 0.190  | 0.794  | 1.203 | 1.140 |
| winter21 | sarima_s1-<br>noTransform  | 4          | 2714.2 | 3710.5 | 0.349  | 0.571  | 1.276 | 1.081 |
| winter21 | sarima_s1-4root            | 4          | 2956.2 | 4062.1 | 0.302  | 0.524  | 1.390 | 1.183 |

| wave  | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| alpha | THieF_2wk-4root            | 1          | 8.843  | 13.759 | 0.527  | 0.941  | 0.548 | 0.815 |
| alpha | THieF_1wk-4root            | 1          | 9.177  | 14.301 | 0.515  | 0.934  | 0.569 | 0.847 |
| alpha | sarima_s1-4root            | 1          | 9.545  | 14.894 | 0.523  | 0.938  | 0.592 | 0.882 |
| alpha | sarima_s7-4root            | 1          | 9.604  | 15.011 | 0.505  | 0.930  | 0.595 | 0.889 |
| alpha | THieF_3wk-4root            | 1          | 10.049 | 15.818 | 0.496  | 0.930  | 0.623 | 0.937 |
| alpha | THieF_1wk-<br>noTransform  | 1          | 10.812 | 14.710 | 0.771  | 0.983  | 0.670 | 0.871 |
| alpha | THieF_2wk-<br>noTransform  | 1          | 10.893 | 14.592 | 0.773  | 0.986  | 0.675 | 0.864 |
| alpha | sarima_s7-<br>noTransform  | 1          | 11.009 | 14.817 | 0.765  | 0.980  | 0.683 | 0.878 |
| alpha | THieF_4wk-4root            | 1          | 11.300 | 17.621 | 0.462  | 0.907  | 0.701 | 1.044 |
| alpha | sarima_s1-<br>noTransform  | 1          | 11.755 | 15.743 | 0.773  | 0.981  | 0.729 | 0.933 |
| alpha | THieF_3wk-<br>noTransform  | 1          | 12.310 | 18.759 | 0.705  | 0.985  | 0.763 | 1.111 |
| alpha | THieF_6wk-4root            | 1          | 12.408 | 19.609 | 0.422  | 0.890  | 0.769 | 1.162 |
| alpha | THieF_4wk-<br>noTransform  | 1          | 13.722 | 21.687 | 0.646  | 0.986  | 0.851 | 1.285 |
| alpha | THieF_8wk-<br>noTransform  | 1          | 14.565 | 22.493 | 0.636  | 0.979  | 0.903 | 1.333 |
| alpha | THieF_8wk-4root            | 1          | 14.931 | 23.120 | 0.380  | 0.834  | 0.926 | 1.370 |
| alpha | THieF_6wk-<br>noTransform  | 1          | 15.423 | 25.717 | 0.539  | 0.987  | 0.956 | 1.524 |
| alpha | COVIDhub-baseline          | 1          | 16.131 | 16.879 | 0.948  | 0.994  | 1.000 | 1.000 |
| alpha | THieF_12wk-4root           | 1          | 22.950 | 32.795 | 0.324  | 0.771  | 1.423 | 1.943 |
| alpha | THieF_12wk-<br>noTransform | 1          | 27.292 | 44.228 | 0.300  | 0.893  | 1.692 | 2.620 |
| alpha | THieF_2wk-4root            | 2          | 13.446 | 20.858 | 0.492  | 0.931  | 0.604 | 0.885 |
| alpha | THieF_1wk-4root            | 2          | 13.592 | 21.198 | 0.511  | 0.934  | 0.610 | 0.899 |
| alpha | sarima_s7-4root            | 2          | 14.048 | 21.702 | 0.508  | 0.927  | 0.631 | 0.920 |
| alpha | sarima_s1-4root            | 2          | 14.060 | 21.658 | 0.511  | 0.929  | 0.631 | 0.918 |
| alpha | THieF_3wk-4root            | 2          | 14.568 | 22.744 | 0.480  | 0.927  | 0.654 | 0.965 |
| alpha | THieF_4wk-4root            | 2          | 15.271 | 23.796 | 0.467  | 0.925  | 0.686 | 1.009 |
| alpha | THieF_1wk-<br>noTransform  | 2          | 15.693 | 22.077 | 0.774  | 0.981  | 0.705 | 0.936 |
| alpha | THieF_2wk-<br>noTransform  | 2          | 15.735 | 21.547 | 0.769  | 0.985  | 0.707 | 0.914 |
| alpha | sarima_s7-<br>noTransform  | 2          | 15.739 | 21.478 | 0.779  | 0.977  | 0.707 | 0.911 |
| alpha | THieF_6wk-4root            | 2          | 15.797 | 25.077 | 0.437  | 0.924  | 0.709 | 1.063 |
| alpha | sarima_s1-<br>noTransform  | 2          | 16.440 | 22.253 | 0.776  | 0.979  | 0.738 | 0.944 |
| alpha | THieF_3wk-<br>noTransform  | 2          | 16.789 | 25.414 | 0.725  | 0.984  | 0.754 | 1.078 |
| alpha | THieF_8wk-4root            | 2          | 17.582 | 27.663 | 0.412  | 0.894  | 0.790 | 1.173 |
| alpha | THieF_8wk-<br>noTransform  | 2          | 17.699 | 26.440 | 0.689  | 0.987  | 0.795 | 1.121 |
| alpha | THieF_4wk-<br>noTransform  | 2          | 17.776 | 27.795 | 0.686  | 0.986  | 0.798 | 1.179 |
| alpha | THieF_6wk-<br>noTransform  | 2          | 19.425 | 31.827 | 0.606  | 0.988  | 0.872 | 1.350 |

| wave  | model                      | horizon_wk | wis     | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|---------|--------|--------|--------|-------|-------|
| alpha | COVIDhub-baseline          | 2          | 22.267  | 23.580 | 0.957  | 0.995  | 1.000 | 1.000 |
| alpha | THieF_12wk-<br>noTransform | 2          | 29.300  | 49.292 | 0.369  | 0.963  | 1.316 | 2.090 |
| alpha | THieF_12wk-4root           | 2          | 38.024  | 36.365 | 0.369  | 0.840  | 1.708 | 1.542 |
| alpha | THieF_1wk-4root            | 3          | 19.977  | 30.494 | 0.482  | 0.922  | 0.714 | 0.958 |
| alpha | THieF_2wk-4root            | 3          | 20.102  | 30.443 | 0.461  | 0.922  | 0.719 | 0.957 |
| alpha | sarima_s1-4root            | 3          | 20.243  | 30.187 | 0.486  | 0.910  | 0.724 | 0.949 |
| alpha | sarima_s7-4root            | 3          | 20.256  | 30.722 | 0.488  | 0.919  | 0.724 | 0.966 |
| alpha | THieF_4wk-4root            | 3          | 21.407  | 32.617 | 0.452  | 0.921  | 0.765 | 1.025 |
| alpha | THieF_6wk-4root            | 3          | 21.435  | 33.299 | 0.428  | 0.921  | 0.766 | 1.047 |
| alpha | THieF_3wk-4root            | 3          | 21.517  | 32.763 | 0.444  | 0.900  | 0.769 | 1.030 |
| alpha | sarima_s7-<br>noTransform  | 3          | 21.721  | 30.684 | 0.758  | 0.974  | 0.777 | 0.964 |
| alpha | THieF_1wk-<br>noTransform  | 3          | 21.931  | 31.984 | 0.748  | 0.977  | 0.784 | 1.005 |
| alpha | sarima_s1-<br>noTransform  | 3          | 21.958  | 30.430 | 0.738  | 0.974  | 0.785 | 0.956 |
| alpha | THieF_2wk-<br>noTransform  | 3          | 22.078  | 31.458 | 0.733  | 0.980  | 0.789 | 0.989 |
| alpha | THieF_8wk-<br>noTransform  | 3          | 22.587  | 34.065 | 0.689  | 0.984  | 0.808 | 1.071 |
| alpha | THieF_8wk-4root            | 3          | 22.821  | 35.565 | 0.402  | 0.914  | 0.816 | 1.118 |
| alpha | THieF_4wk-<br>noTransform  | 3          | 23.146  | 35.988 | 0.695  | 0.978  | 0.828 | 1.131 |
| alpha | THieF_3wk-<br>noTransform  | 3          | 23.577  | 36.486 | 0.677  | 0.977  | 0.843 | 1.147 |
| alpha | THieF_6wk-<br>noTransform  | 3          | 24.893  | 40.185 | 0.618  | 0.983  | 0.890 | 1.263 |
| alpha | COVIDhub-baseline          | 3          | 27.968  | 31.818 | 0.948  | 0.993  | 1.000 | 1.000 |
| alpha | THieF_12wk-<br>noTransform | 3          | 34.831  | 58.662 | 0.384  | 0.969  | 1.245 | 1.844 |
| alpha | THieF_12wk-4root           | 3          | 126.715 | 44.356 | 0.365  | 0.859  | 4.531 | 1.394 |
| alpha | sarima_s1-4root            | 4          | 28.477  | 40.814 | 0.452  | 0.874  | 0.830 | 0.972 |
| alpha | THieF_1wk-4root            | 4          | 28.491  | 42.006 | 0.449  | 0.895  | 0.830 | 1.000 |
| alpha | THieF_8wk-<br>noTransform  | 4          | 28.501  | 43.156 | 0.682  | 0.975  | 0.830 | 1.027 |
| alpha | sarima_s7-4root            | 4          | 28.528  | 41.710 | 0.454  | 0.893  | 0.831 | 0.993 |
| alpha | THieF_2wk-4root            | 4          | 28.815  | 42.263 | 0.425  | 0.888  | 0.839 | 1.006 |
| alpha | sarima_s1-<br>noTransform  | 4          | 28.835  | 40.658 | 0.703  | 0.963  | 0.840 | 0.968 |
| alpha | sarima_s7-<br>noTransform  | 4          | 29.059  | 41.948 | 0.729  | 0.963  | 0.847 | 0.999 |
| alpha | THieF_6wk-4root            | 4          | 29.188  | 43.894 | 0.403  | 0.899  | 0.850 | 1.045 |
| alpha | THieF_2wk-<br>noTransform  | 4          | 29.331  | 42.151 | 0.701  | 0.968  | 0.854 | 1.004 |
| alpha | THieF_1wk-<br>noTransform  | 4          | 29.463  | 43.731 | 0.723  | 0.967  | 0.858 | 1.041 |
| alpha | THieF_4wk-4root            | 4          | 29.935  | 43.877 | 0.416  | 0.892  | 0.872 | 1.045 |
| alpha | THieF_8wk-4root            | 4          | 30.176  | 45.634 | 0.396  | 0.905  | 0.879 | 1.086 |
| alpha | THieF_4wk-<br>noTransform  | 4          | 30.303  | 46.903 | 0.662  | 0.964  | 0.883 | 1.117 |

| wave  | model                      | horizon_wk | wis     | mae    | cov_50 | cov_95 | rwis   | rmae  |
|-------|----------------------------|------------|---------|--------|--------|--------|--------|-------|
| alpha | THieF_6wk-<br>noTransform  | 4          | 30.499  | 47.889 | 0.637  | 0.973  | 0.889  | 1.140 |
| alpha | THieF_3wk-4root            | 4          | 30.963  | 45.743 | 0.401  | 0.880  | 0.902  | 1.089 |
| alpha | THieF_3wk-<br>noTransform  | 4          | 32.277  | 51.096 | 0.623  | 0.969  | 0.940  | 1.217 |
| alpha | COVIDhub-baseline          | 4          | 34.326  | 42.002 | 0.930  | 0.991  | 1.000  | 1.000 |
| alpha | THieF_12wk-<br>noTransform | 4          | 43.033  | 71.829 | 0.373  | 0.964  | 1.254  | 1.710 |
| alpha | THieF_12wk-4root           | 4          | 482.735 | 56.833 | 0.348  | 0.847  | 14.063 | 1.353 |
| delta | THieF_2wk-4root            | 1          | 17.117  | 26.578 | 0.539  | 0.945  | 0.695  | 0.735 |
| delta | THieF_1wk-4root            | 1          | 17.713  | 27.689 | 0.532  | 0.943  | 0.719  | 0.766 |
| delta | THieF_2wk-<br>noTransform  | 1          | 19.117  | 27.527 | 0.531  | 0.897  | 0.776  | 0.761 |
| delta | sarima_s1-4root            | 1          | 19.732  | 30.568 | 0.503  | 0.930  | 0.801  | 0.846 |
| delta | THieF_3wk-4root            | 1          | 20.002  | 30.483 | 0.510  | 0.930  | 0.812  | 0.843 |
| delta | THieF_1wk-<br>noTransform  | 1          | 20.269  | 29.576 | 0.504  | 0.886  | 0.823  | 0.818 |
| delta | THieF_3wk-<br>noTransform  | 1          | 20.860  | 30.399 | 0.511  | 0.884  | 0.847  | 0.841 |
| delta | sarima_s7-4root            | 1          | 21.094  | 32.455 | 0.498  | 0.920  | 0.856  | 0.898 |
| delta | THieF_4wk-4root            | 1          | 21.785  | 32.633 | 0.507  | 0.916  | 0.884  | 0.903 |
| delta | sarima_s7-<br>noTransform  | 1          | 22.277  | 32.170 | 0.479  | 0.867  | 0.904  | 0.890 |
| delta | THieF_6wk-4root            | 1          | 22.383  | 33.376 | 0.489  | 0.917  | 0.909  | 0.923 |
| delta | THieF_6wk-<br>noTransform  | 1          | 22.876  | 33.136 | 0.500  | 0.883  | 0.929  | 0.917 |
| delta | sarima_s1-<br>noTransform  | 1          | 23.144  | 33.894 | 0.478  | 0.874  | 0.939  | 0.938 |
| delta | THieF_4wk-<br>noTransform  | 1          | 23.584  | 33.893 | 0.495  | 0.872  | 0.957  | 0.937 |
| delta | COVIDhub-baseline          | 1          | 24.637  | 36.152 | 0.716  | 0.972  | 1.000  | 1.000 |
| delta | THieF_8wk-<br>noTransform  | 1          | 24.923  | 35.858 | 0.485  | 0.871  | 1.012  | 0.992 |
| delta | THieF_12wk-4root           | 1          | 24.931  | 36.375 | 0.494  | 0.906  | 1.012  | 1.006 |
| delta | THieF_8wk-4root            | 1          | 26.010  | 38.121 | 0.461  | 0.890  | 1.056  | 1.054 |
| delta | THieF_12wk-<br>noTransform | 1          | 28.228  | 39.426 | 0.476  | 0.839  | 1.146  | 1.091 |
| delta | THieF_2wk-4root            | 2          | 28.151  | 43.242 | 0.500  | 0.921  | 0.740  | 0.762 |
| delta | THieF_6wk-4root            | 2          | 28.642  | 42.877 | 0.512  | 0.923  | 0.752  | 0.755 |
| delta | THieF_1wk-4root            | 2          | 28.832  | 44.944 | 0.490  | 0.922  | 0.757  | 0.792 |
| delta | THieF_2wk-<br>noTransform  | 2          | 29.422  | 42.924 | 0.508  | 0.876  | 0.773  | 0.756 |
| delta | THieF_3wk-4root            | 2          | 30.439  | 46.335 | 0.501  | 0.906  | 0.800  | 0.816 |
| delta | THieF_6wk-<br>noTransform  | 2          | 30.731  | 45.131 | 0.518  | 0.889  | 0.807  | 0.795 |
| delta | THieF_12wk-4root           | 2          | 31.321  | 45.905 | 0.499  | 0.914  | 0.823  | 0.809 |
| delta | THieF_8wk-4root            | 2          | 31.400  | 46.685 | 0.485  | 0.908  | 0.825  | 0.823 |
| delta | THieF_4wk-4root            | 2          | 31.436  | 46.948 | 0.493  | 0.897  | 0.826  | 0.827 |
| delta | THieF_8wk-<br>noTransform  | 2          | 31.456  | 46.398 | 0.497  | 0.877  | 0.826  | 0.818 |
| delta | THieF_3wk-<br>noTransform  | 2          | 31.608  | 45.968 | 0.494  | 0.870  | 0.830  | 0.810 |

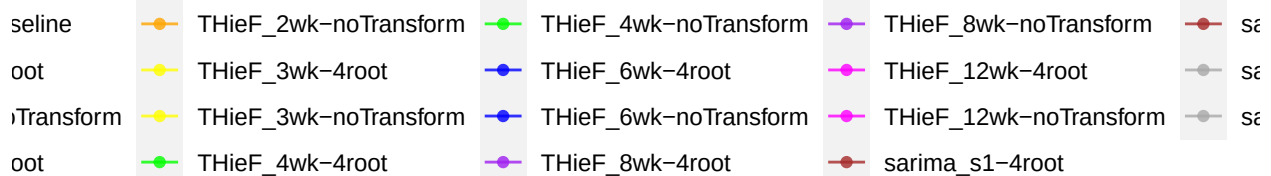
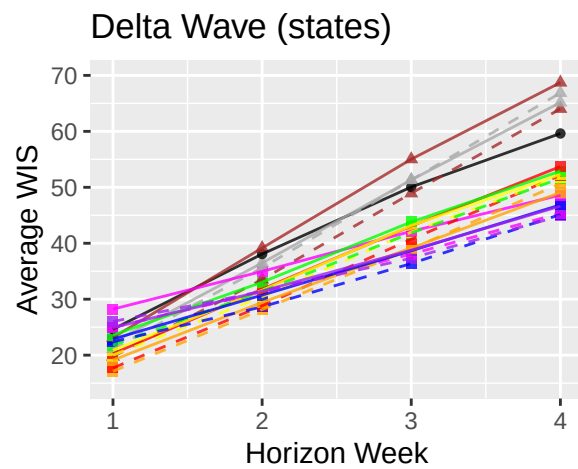
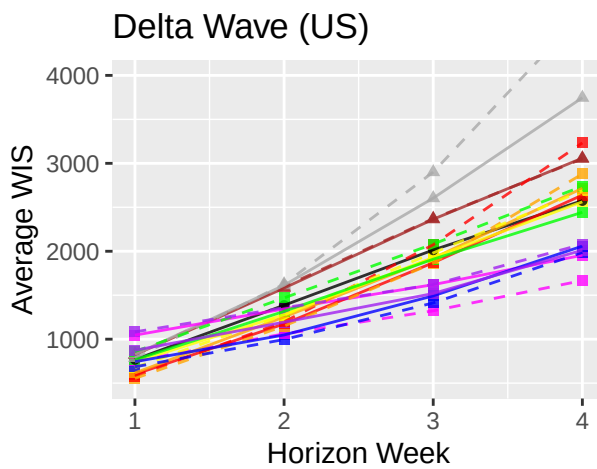
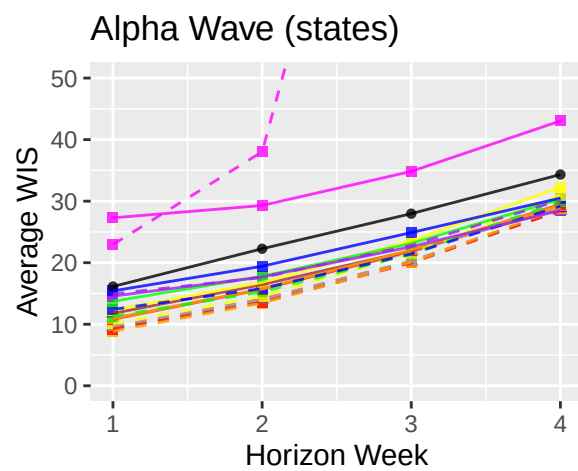
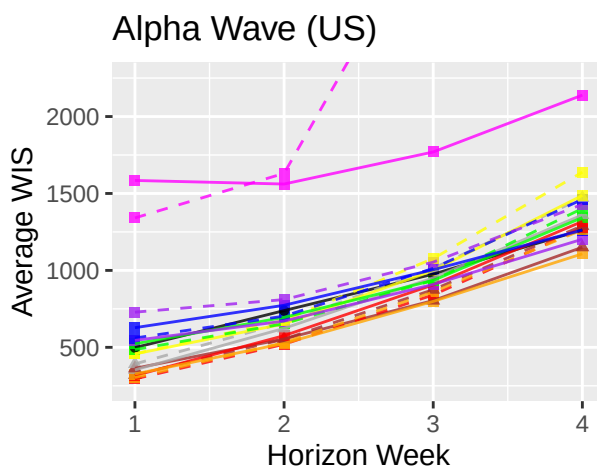
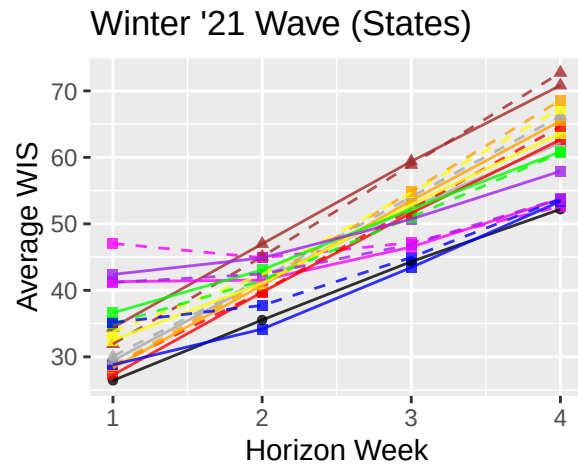
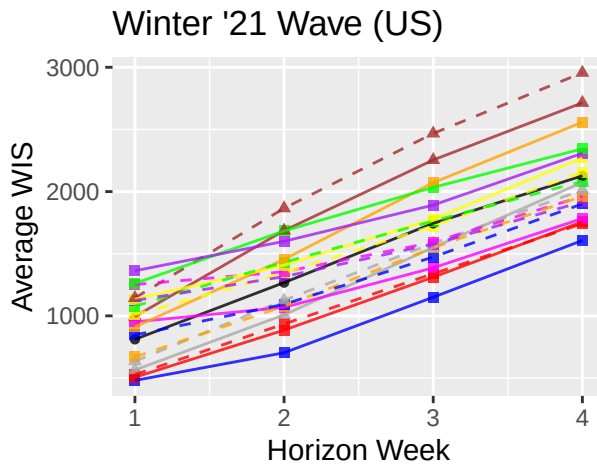
| wave  | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|-------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| delta | THieF_1wk-<br>noTransform  | 2          | 31.783 | 46.745 | 0.483  | 0.868  | 0.835 | 0.824 |
| delta | THieF_4wk-<br>noTransform  | 2          | 33.040 | 48.561 | 0.481  | 0.864  | 0.868 | 0.856 |
| delta | sarima_s1-4root            | 2          | 33.524 | 51.293 | 0.448  | 0.893  | 0.881 | 0.904 |
| delta | THieF_12wk-<br>noTransform | 2          | 34.973 | 49.279 | 0.497  | 0.854  | 0.919 | 0.868 |
| delta | sarima_s7-4root            | 2          | 35.617 | 53.896 | 0.450  | 0.894  | 0.936 | 0.950 |
| delta | sarima_s7-<br>noTransform  | 2          | 36.488 | 52.231 | 0.457  | 0.844  | 0.959 | 0.920 |
| delta | COVIDhub-baseline          | 2          | 38.067 | 56.755 | 0.723  | 0.972  | 1.000 | 1.000 |
| delta | sarima_s1-<br>noTransform  | 2          | 39.151 | 56.855 | 0.436  | 0.841  | 1.028 | 1.002 |
| delta | THieF_6wk-4root            | 3          | 36.372 | 53.911 | 0.508  | 0.914  | 0.727 | 0.714 |
| delta | THieF_8wk-4root            | 3          | 37.330 | 55.151 | 0.505  | 0.910  | 0.746 | 0.730 |
| delta | THieF_12wk-4root           | 3          | 38.068 | 55.041 | 0.495  | 0.911  | 0.761 | 0.729 |
| delta | THieF_6wk-<br>noTransform  | 3          | 38.661 | 57.250 | 0.511  | 0.879  | 0.773 | 0.758 |
| delta | THieF_8wk-<br>noTransform  | 3          | 38.775 | 57.642 | 0.500  | 0.881  | 0.775 | 0.763 |
| delta | THieF_2wk-4root            | 3          | 38.991 | 59.030 | 0.495  | 0.903  | 0.779 | 0.782 |
| delta | THieF_2wk-<br>noTransform  | 3          | 39.153 | 57.544 | 0.503  | 0.870  | 0.782 | 0.762 |
| delta | THieF_1wk-4root            | 3          | 40.707 | 62.441 | 0.470  | 0.895  | 0.813 | 0.827 |
| delta | THieF_4wk-4root            | 3          | 41.824 | 62.384 | 0.478  | 0.884  | 0.836 | 0.826 |
| delta | THieF_3wk-4root            | 3          | 41.997 | 63.577 | 0.483  | 0.884  | 0.839 | 0.842 |
| delta | THieF_12wk-<br>noTransform | 3          | 42.134 | 58.490 | 0.493  | 0.859  | 0.842 | 0.774 |
| delta | THieF_1wk-<br>noTransform  | 3          | 43.072 | 63.570 | 0.460  | 0.857  | 0.861 | 0.842 |
| delta | THieF_3wk-<br>noTransform  | 3          | 43.159 | 62.801 | 0.472  | 0.857  | 0.862 | 0.832 |
| delta | THieF_4wk-<br>noTransform  | 3          | 43.804 | 64.424 | 0.461  | 0.853  | 0.875 | 0.853 |
| delta | sarima_s1-4root            | 3          | 48.925 | 73.609 | 0.421  | 0.864  | 0.978 | 0.975 |
| delta | COVIDhub-baseline          | 3          | 50.044 | 75.526 | 0.705  | 0.965  | 1.000 | 1.000 |
| delta | sarima_s7-4root            | 3          | 51.297 | 75.911 | 0.431  | 0.866  | 1.025 | 1.005 |
| delta | sarima_s7-<br>noTransform  | 3          | 51.326 | 72.880 | 0.440  | 0.825  | 1.026 | 0.965 |
| delta | sarima_s1-<br>noTransform  | 3          | 55.021 | 79.067 | 0.405  | 0.817  | 1.099 | 1.047 |
| delta | THieF_8wk-4root            | 4          | 44.907 | 66.666 | 0.487  | 0.902  | 0.753 | 0.733 |
| delta | THieF_6wk-4root            | 4          | 45.143 | 67.184 | 0.482  | 0.901  | 0.757 | 0.738 |
| delta | THieF_12wk-4root           | 4          | 45.372 | 64.617 | 0.474  | 0.902  | 0.761 | 0.710 |
| delta | THieF_8wk-<br>noTransform  | 4          | 46.640 | 70.382 | 0.483  | 0.875  | 0.782 | 0.773 |
| delta | THieF_6wk-<br>noTransform  | 4          | 46.858 | 70.722 | 0.472  | 0.874  | 0.786 | 0.777 |
| delta | THieF_12wk-<br>noTransform | 4          | 48.614 | 67.409 | 0.486  | 0.860  | 0.815 | 0.741 |
| delta | THieF_2wk-<br>noTransform  | 4          | 49.075 | 72.591 | 0.476  | 0.868  | 0.823 | 0.798 |

| wave     | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|----------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| delta    | THieF_2wk-4root            | 4          | 50.648 | 76.480 | 0.465  | 0.886  | 0.850 | 0.840 |
| delta    | THieF_4wk-4root            | 4          | 51.612 | 77.311 | 0.463  | 0.877  | 0.866 | 0.850 |
| delta    | THieF_3wk-4root            | 4          | 51.841 | 78.166 | 0.478  | 0.877  | 0.870 | 0.859 |
| delta    | THieF_1wk-4root            | 4          | 52.152 | 79.195 | 0.455  | 0.875  | 0.875 | 0.870 |
| delta    | THieF_3wk-<br>noTransform  | 4          | 52.315 | 77.297 | 0.456  | 0.856  | 0.877 | 0.849 |
| delta    | THieF_4wk-<br>noTransform  | 4          | 52.897 | 78.704 | 0.440  | 0.844  | 0.887 | 0.865 |
| delta    | THieF_1wk-<br>noTransform  | 4          | 53.715 | 79.799 | 0.446  | 0.843  | 0.901 | 0.877 |
| delta    | COVIDhub-baseline          | 4          | 59.620 | 90.998 | 0.692  | 0.965  | 1.000 | 1.000 |
| delta    | sarima_s1-4root            | 4          | 64.026 | 95.362 | 0.400  | 0.845  | 1.074 | 1.048 |
| delta    | sarima_s7-<br>noTransform  | 4          | 65.158 | 92.398 | 0.427  | 0.811  | 1.093 | 1.015 |
| delta    | sarima_s7-4root            | 4          | 66.871 | 97.487 | 0.426  | 0.853  | 1.122 | 1.071 |
| delta    | sarima_s1-<br>noTransform  | 4          | 68.721 | 98.267 | 0.378  | 0.797  | 1.153 | 1.080 |
| winter21 | COVIDhub-baseline          | 1          | 26.445 | 37.213 | 0.783  | 0.977  | 1.000 | 1.000 |
| winter21 | THieF_1wk-<br>noTransform  | 1          | 27.270 | 39.076 | 0.514  | 0.864  | 1.031 | 1.050 |
| winter21 | THieF_2wk-<br>noTransform  | 1          | 28.323 | 40.803 | 0.493  | 0.860  | 1.071 | 1.096 |
| winter21 | THieF_1wk-4root            | 1          | 28.616 | 42.525 | 0.525  | 0.921  | 1.082 | 1.143 |
| winter21 | THieF_2wk-4root            | 1          | 28.651 | 44.121 | 0.500  | 0.925  | 1.083 | 1.186 |
| winter21 | THieF_6wk-<br>noTransform  | 1          | 28.802 | 40.390 | 0.525  | 0.881  | 1.089 | 1.085 |
| winter21 | sarima_s7-<br>noTransform  | 1          | 29.406 | 41.562 | 0.501  | 0.855  | 1.112 | 1.117 |
| winter21 | sarima_s7-4root            | 1          | 30.056 | 43.792 | 0.511  | 0.909  | 1.137 | 1.177 |
| winter21 | sarima_s1-4root            | 1          | 31.931 | 46.848 | 0.486  | 0.903  | 1.207 | 1.259 |
| winter21 | THieF_3wk-<br>noTransform  | 1          | 32.516 | 47.244 | 0.428  | 0.845  | 1.230 | 1.270 |
| winter21 | THieF_3wk-4root            | 1          | 33.797 | 50.763 | 0.458  | 0.904  | 1.278 | 1.364 |
| winter21 | sarima_s1-<br>noTransform  | 1          | 34.411 | 48.460 | 0.473  | 0.849  | 1.301 | 1.302 |
| winter21 | THieF_4wk-4root            | 1          | 34.972 | 52.524 | 0.430  | 0.884  | 1.322 | 1.411 |
| winter21 | THieF_6wk-4root            | 1          | 35.142 | 51.733 | 0.429  | 0.886  | 1.329 | 1.390 |
| winter21 | THieF_4wk-<br>noTransform  | 1          | 36.620 | 52.933 | 0.391  | 0.826  | 1.385 | 1.422 |
| winter21 | THieF_8wk-4root            | 1          | 41.175 | 59.110 | 0.384  | 0.829  | 1.557 | 1.588 |
| winter21 | THieF_12wk-<br>noTransform | 1          | 41.303 | 57.208 | 0.385  | 0.829  | 1.562 | 1.537 |
| winter21 | THieF_8wk-<br>noTransform  | 1          | 42.393 | 60.436 | 0.324  | 0.805  | 1.603 | 1.624 |
| winter21 | THieF_12wk-4root           | 1          | 47.050 | 66.133 | 0.331  | 0.788  | 1.779 | 1.777 |
| winter21 | THieF_6wk-<br>noTransform  | 2          | 34.175 | 50.498 | 0.535  | 0.901  | 0.962 | 1.014 |
| winter21 | COVIDhub-baseline          | 2          | 35.542 | 49.816 | 0.836  | 0.990  | 1.000 | 1.000 |
| winter21 | THieF_6wk-4root            | 2          | 37.730 | 58.242 | 0.453  | 0.918  | 1.062 | 1.169 |
| winter21 | THieF_1wk-<br>noTransform  | 2          | 39.715 | 57.621 | 0.495  | 0.856  | 1.117 | 1.157 |
| winter21 | THieF_1wk-4root            | 2          | 39.799 | 61.270 | 0.469  | 0.914  | 1.120 | 1.230 |

| wave     | model                      | horizon_wk | wis    | mae    | cov_50 | cov_95 | rwis  | rmae  |
|----------|----------------------------|------------|--------|--------|--------|--------|-------|-------|
| winter21 | THieF_2wk-<br>noTransform  | 2          | 40.771 | 59.161 | 0.469  | 0.842  | 1.147 | 1.188 |
| winter21 | THieF_3wk-<br>noTransform  | 2          | 41.058 | 61.256 | 0.437  | 0.844  | 1.155 | 1.230 |
| winter21 | THieF_2wk-4root            | 2          | 41.250 | 64.381 | 0.443  | 0.905  | 1.161 | 1.292 |
| winter21 | sarima_s7-<br>noTransform  | 2          | 41.464 | 59.614 | 0.487  | 0.846  | 1.167 | 1.197 |
| winter21 | THieF_4wk-4root            | 2          | 41.483 | 63.855 | 0.432  | 0.897  | 1.167 | 1.282 |
| winter21 | THieF_12wk-<br>noTransform | 2          | 41.544 | 61.768 | 0.425  | 0.882  | 1.169 | 1.240 |
| winter21 | sarima_s7-4root            | 2          | 41.752 | 62.716 | 0.465  | 0.899  | 1.175 | 1.259 |
| winter21 | THieF_3wk-4root            | 2          | 42.361 | 65.908 | 0.432  | 0.905  | 1.192 | 1.323 |
| winter21 | THieF_8wk-4root            | 2          | 42.455 | 64.775 | 0.396  | 0.886  | 1.195 | 1.300 |
| winter21 | THieF_4wk-<br>noTransform  | 2          | 43.072 | 63.412 | 0.434  | 0.842  | 1.212 | 1.273 |
| winter21 | THieF_8wk-<br>noTransform  | 2          | 44.874 | 67.923 | 0.344  | 0.855  | 1.263 | 1.363 |
| winter21 | THieF_12wk-4root           | 2          | 44.902 | 67.133 | 0.380  | 0.864  | 1.263 | 1.348 |
| winter21 | sarima_s1-4root            | 2          | 45.062 | 67.606 | 0.434  | 0.879  | 1.268 | 1.357 |
| winter21 | sarima_s1-<br>noTransform  | 2          | 46.990 | 67.836 | 0.460  | 0.827  | 1.322 | 1.362 |
| winter21 | THieF_6wk-<br>noTransform  | 3          | 43.440 | 65.034 | 0.511  | 0.890  | 0.980 | 1.016 |
| winter21 | COVIDhub-baseline          | 3          | 44.312 | 64.019 | 0.840  | 0.992  | 1.000 | 1.000 |
| winter21 | THieF_6wk-4root            | 3          | 44.956 | 70.110 | 0.459  | 0.913  | 1.015 | 1.095 |
| winter21 | THieF_12wk-<br>noTransform | 3          | 46.517 | 71.337 | 0.425  | 0.886  | 1.050 | 1.114 |
| winter21 | THieF_8wk-4root            | 3          | 46.882 | 73.246 | 0.401  | 0.892  | 1.058 | 1.144 |
| winter21 | THieF_12wk-4root           | 3          | 47.184 | 72.416 | 0.397  | 0.883  | 1.065 | 1.131 |
| winter21 | THieF_8wk-<br>noTransform  | 3          | 50.706 | 77.684 | 0.371  | 0.850  | 1.144 | 1.213 |
| winter21 | THieF_4wk-4root            | 3          | 51.002 | 78.241 | 0.431  | 0.881  | 1.151 | 1.222 |
| winter21 | THieF_1wk-<br>noTransform  | 3          | 51.619 | 75.172 | 0.488  | 0.845  | 1.165 | 1.174 |
| winter21 | THieF_1wk-4root            | 3          | 52.188 | 80.982 | 0.435  | 0.898  | 1.178 | 1.265 |
| winter21 | THieF_4wk-<br>noTransform  | 3          | 52.223 | 76.131 | 0.460  | 0.832  | 1.179 | 1.189 |
| winter21 | sarima_s7-<br>noTransform  | 3          | 52.425 | 76.252 | 0.479  | 0.843  | 1.183 | 1.191 |
| winter21 | THieF_3wk-<br>noTransform  | 3          | 52.885 | 78.854 | 0.425  | 0.834  | 1.193 | 1.232 |
| winter21 | THieF_2wk-<br>noTransform  | 3          | 53.500 | 77.562 | 0.453  | 0.832  | 1.207 | 1.212 |
| winter21 | sarima_s7-4root            | 3          | 54.054 | 82.641 | 0.431  | 0.890  | 1.220 | 1.291 |
| winter21 | THieF_3wk-4root            | 3          | 54.705 | 84.514 | 0.416  | 0.879  | 1.235 | 1.320 |
| winter21 | THieF_2wk-4root            | 3          | 54.851 | 84.772 | 0.420  | 0.879  | 1.238 | 1.324 |
| winter21 | sarima_s1-4root            | 3          | 58.883 | 89.083 | 0.403  | 0.849  | 1.329 | 1.391 |
| winter21 | sarima_s1-<br>noTransform  | 3          | 59.403 | 86.210 | 0.441  | 0.810  | 1.341 | 1.347 |
| winter21 | COVIDhub-baseline          | 4          | 52.196 | 76.859 | 0.839  | 0.990  | 1.000 | 1.000 |
| winter21 | THieF_12wk-4root           | 4          | 52.631 | 80.627 | 0.406  | 0.881  | 1.008 | 1.049 |

| wave     | model                      | horizon_wk | wis    | mae     | cov_50 | cov_95 | rwis  | rmae  |
|----------|----------------------------|------------|--------|---------|--------|--------|-------|-------|
| winter21 | THieF_6wk-<br>noTransform  | 4          | 53.396 | 79.066  | 0.500  | 0.871  | 1.023 | 1.029 |
| winter21 | THieF_6wk-4root            | 4          | 53.722 | 82.813  | 0.451  | 0.890  | 1.029 | 1.077 |
| winter21 | THieF_12wk-<br>noTransform | 4          | 53.730 | 82.243  | 0.432  | 0.883  | 1.029 | 1.070 |
| winter21 | THieF_8wk-4root            | 4          | 53.816 | 83.148  | 0.415  | 0.882  | 1.031 | 1.082 |
| winter21 | THieF_8wk-<br>noTransform  | 4          | 57.911 | 88.095  | 0.410  | 0.848  | 1.109 | 1.146 |
| winter21 | THieF_4wk-4root            | 4          | 60.668 | 91.715  | 0.430  | 0.869  | 1.162 | 1.193 |
| winter21 | THieF_4wk-<br>noTransform  | 4          | 60.851 | 88.155  | 0.475  | 0.831  | 1.166 | 1.147 |
| winter21 | sarima_s7-<br>noTransform  | 4          | 62.193 | 91.008  | 0.477  | 0.834  | 1.192 | 1.184 |
| winter21 | THieF_1wk-<br>noTransform  | 4          | 62.660 | 90.964  | 0.482  | 0.836  | 1.200 | 1.184 |
| winter21 | THieF_3wk-<br>noTransform  | 4          | 63.610 | 94.374  | 0.436  | 0.832  | 1.219 | 1.228 |
| winter21 | THieF_1wk-4root            | 4          | 64.630 | 99.459  | 0.425  | 0.873  | 1.238 | 1.294 |
| winter21 | THieF_2wk-<br>noTransform  | 4          | 65.471 | 93.901  | 0.446  | 0.818  | 1.254 | 1.222 |
| winter21 | sarima_s7-4root            | 4          | 66.169 | 101.509 | 0.412  | 0.874  | 1.268 | 1.321 |
| winter21 | THieF_3wk-4root            | 4          | 67.320 | 102.009 | 0.416  | 0.861  | 1.290 | 1.327 |
| winter21 | THieF_2wk-4root            | 4          | 68.582 | 103.751 | 0.414  | 0.858  | 1.314 | 1.350 |
| winter21 | sarima_s1-<br>noTransform  | 4          | 70.817 | 102.300 | 0.432  | 0.798  | 1.357 | 1.331 |
| winter21 | sarima_s1-4root            | 4          | 72.729 | 109.494 | 0.384  | 0.825  | 1.393 | 1.425 |

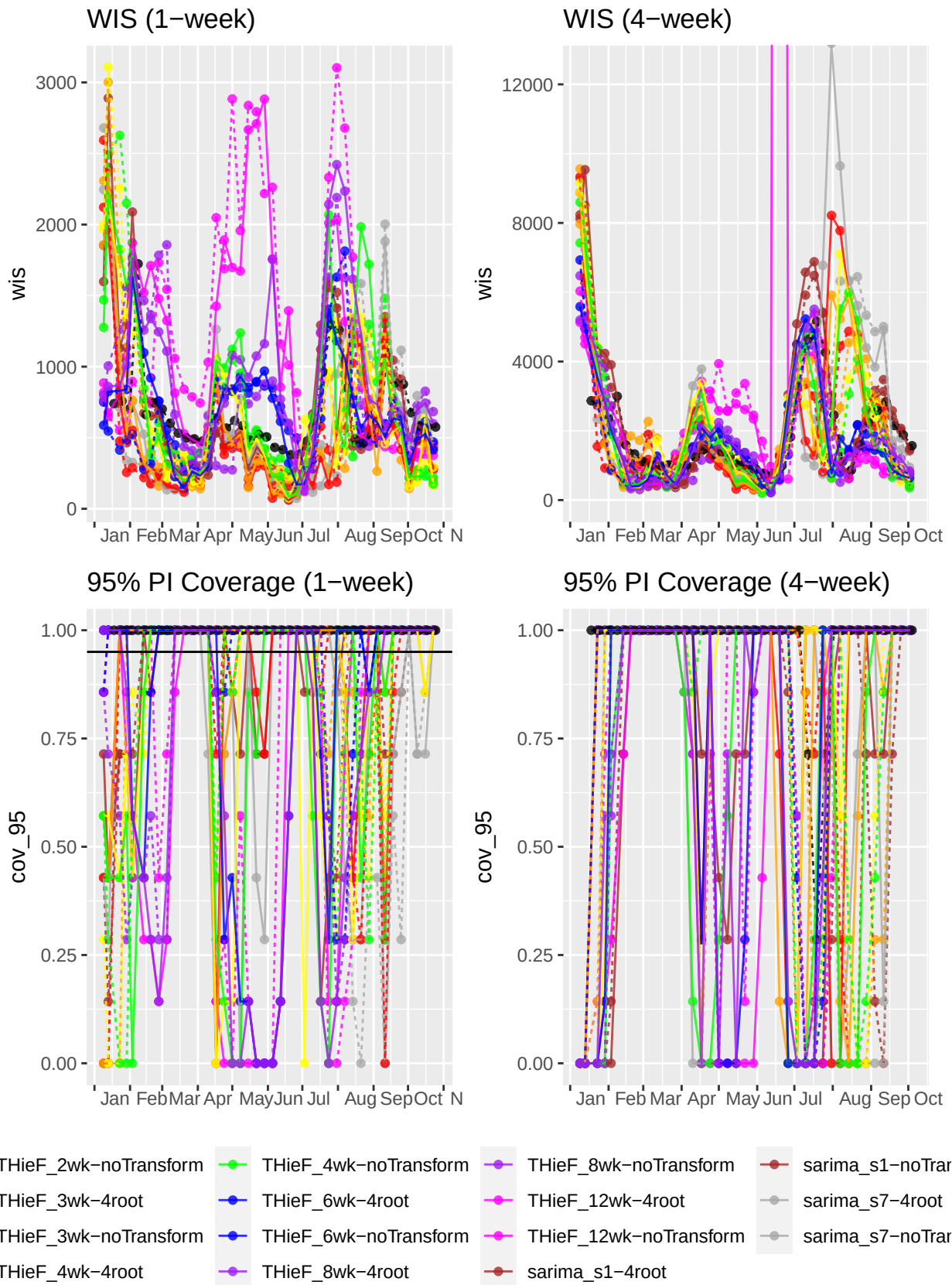




For US National during the alpha wave, the Base\_arima\_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base\_arima\_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8\_arima\_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8\_arima\_4root and Multiple3\_arima\_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw fairly comparable or worse performance compared to the baseline for WIS and MAE. The 50% coverage was better than the baseline for all models, the greatest improvements for the best models, but the 95% coverage could be better or worse for the best models.

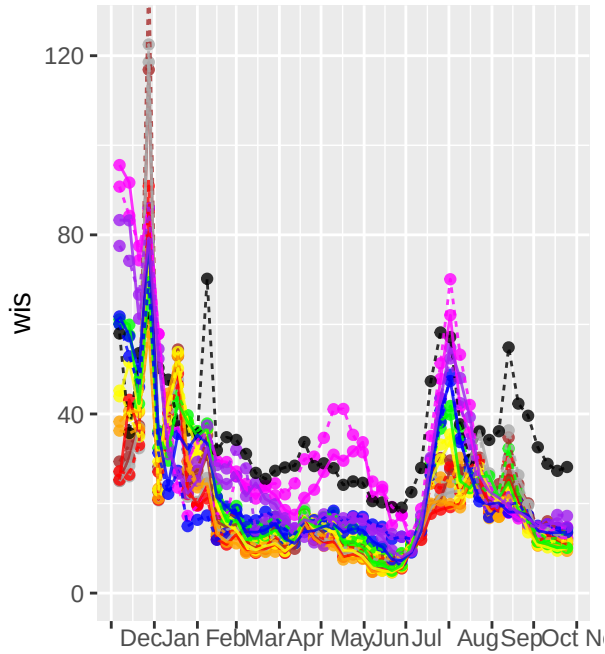
For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base\_arima\_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8\_arima\_noTrans and Base\_arima\_noTrans models, then the Topmost8\_arima\_4root and Multiple3\_arima\_noTrans models, all of which are much better than the Multiple3\_arima\_4root model. During the winter21 wave, the Base\_arima\_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one. Similar patterns to the ones described above for the US national level when compared to the baseline were observed.

By forecast date (week)

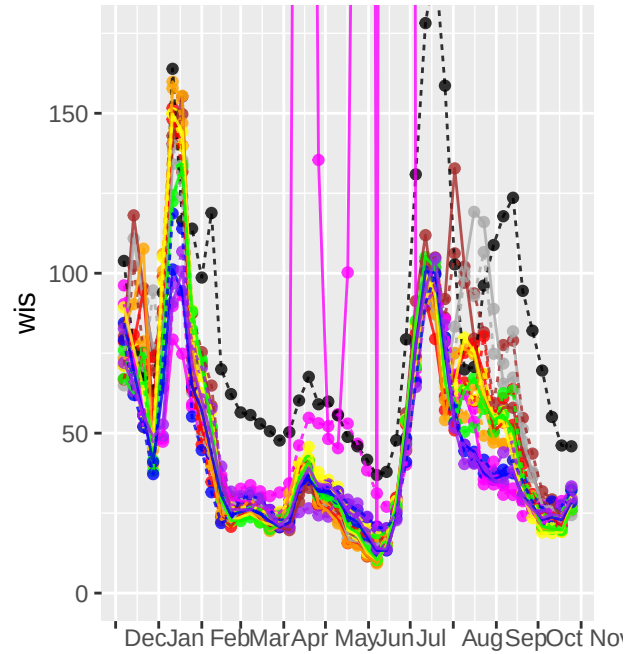


```
## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
## 'summarise()' has grouped output by 'forecast_date', 'model'. You can override
## using the '.groups' argument.
```

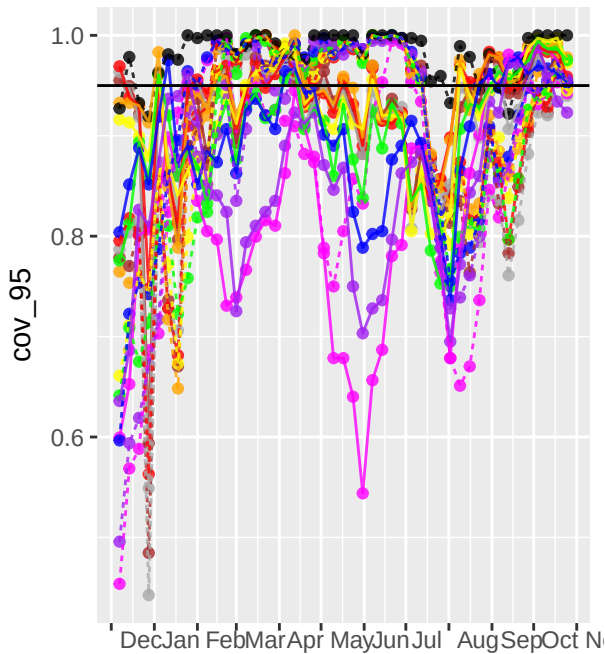
WIS (1-week)



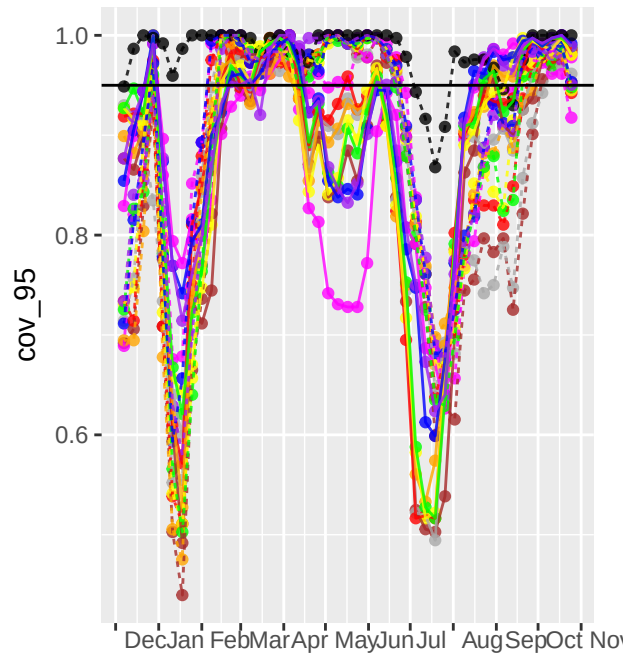
WIS (4-week)



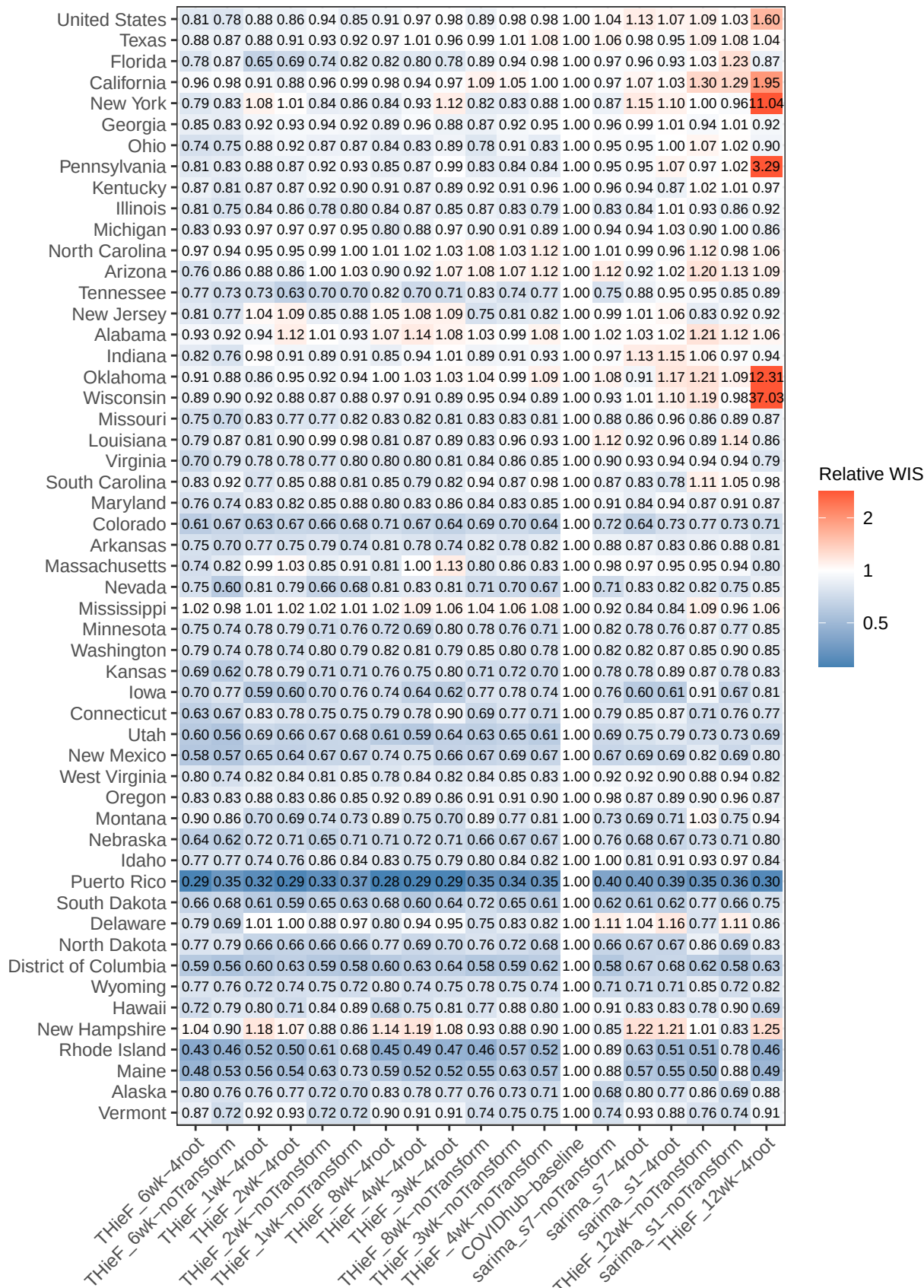
95% PI Coverage (1-week)



95% PI Coverage (4-week)



## Metrics By Location



## Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base\_arima\_noTrans, the Base\_arima\_4root, and the Topmost8\_arima\_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.